

The Role of Crisis Pregnancy Centers in Fertility Decisions*

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Abstract

Crisis pregnancy centers (CPCs) outnumber abortion clinics three to one and have received \$489 million in public funding since Dobbs, yet no causal evidence exists on whether they affect fertility. Because CPCs locate in response to unobserved abortion demand, I construct a simulated instrument predicting counterfactual CPC expansion across a 30-year North and South Carolina county panel. One additional CPC per 10,000 women reduces abortion rates by 7.0 to 14.2 percent. The aggregate pregnancy response is indistinguishable from zero, indicating substitution from abortion to birth, not deterrence. Welfare concerns are sharpest among teenagers, whose CPC-induced births carry higher low-birth-weight rates.

JEL Codes: J12, J13, I12, C15, C36

Keywords: fertility, abortion, birth, crisis pregnancy centers, simulated instruments

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1 Introduction

More than 2,500 crisis pregnancy centers (CPCs) operate in the United States, outnumbering abortion clinics three to one (Jones and Jerman, 2017; Lambert et al., 2025). They counsel pregnant women toward carrying to term, offering pregnancy testing, ultrasound, material support, and community networks (Care Net, 2018). Between 12 and 20 percent of women with a pregnancy history have visited one (Yang et al., 2025). CPCs received at least \$89 million in public funding annually at the end of the analysis period (Kruesi, 2022), and states have directed an additional \$489 million to CPCs since *Dobbs v. Jackson Women’s Health Organization* (CBS News, 2024). Care Net, a national CPC network, reports that its 1,100 affiliated centers prevented 1 in 10 abortions between 2008 and 2017 (677,248 in total) (Care Net, 2018). Despite this scale, no causal evidence exists on whether CPCs affect fertility outcomes.

This paper makes three contributions. First, it provides causal evidence that CPCs reduce the abortion rate, with the offsetting birth response concentrated among women aged 25 and older. Second, it develops an IV strategy that generalizes simulated instruments to settings without exogenous policy variation (Currie and Gruber, 1996). Whereas Currie and Gruber simulate Medicaid eligibility, this paper simulates counterfactual CPC expansion paths. Akin to formula instruments (Borusyak et al., 2025), the approach is portable to settings where the treatment assignment mechanism can be explicitly modeled. In a 30-year county panel from North Carolina and South Carolina (1990–2019), the instrument averages predicted CPC exposure across simulated expansion paths. Monte Carlo experiments confirm that the estimator recovers the parameter of interest even under deliberate misspecification of the entry model. Third, it builds a welfare framework that identifies teens as the subgroup where the cost of abortion-to-birth substitution concentrates. Among teenagers, the marginal births have worse health outcomes than the cell average, and teen mothers face disproportionate downstream economic costs (Kearney and Levine, 2012), making this the subgroup where welfare concerns are most empirically grounded.

Three literatures bear on this question. Supply-side fertility research documents how access restrictions affect abortion and births (Lindo et al., 2020; Myers, 2017; Fischer et al., 2018; Quast et al., 2017; Lu and Slusky, 2019). The abortion result in this paper is comparable in magnitude to clinic closures (Lindo et al., 2020) and to increases in distance to the nearest abortion

provider (Myers, 2017), but the mechanism is demand-side persuasion rather than access restriction. Demand-side fertility intervention has focused on mass media, including MTV's *16 and Pregnant* (Kearney and Levine, 2015), Brazilian telenovelas (La Ferrara et al., 2012), and Indian cable (Jensen and Oster, 2009). In contrast, CPCs counsel pregnant women in person toward a specific decision. They are often difficult for women to distinguish from abortion providers or medical clinics (Swartz et al., 2021). The closest analog for CPCs in the economics of persuasion is DellaVigna and Gentzkow (2010)'s finding that in-person canvassing shifts behavior an order of magnitude more than mass-media exposure. An experimental literature now studies how persuasion, social-norm shifts, and narratives move consequential behavior (Bursztyn et al., 2020; Cantoni et al., 2019; Andre et al., 2025). For that literature, the CPC setting offers an unusual combination of ingredients: the sender's strategy is observable through training materials and mystery-client audits, receivers' priors are partially observable through mission-awareness surveys, and exposure varies quasi-experimentally through the simulated-instrument design.

One additional CPC per 10,000 women reduces abortion rates by 7.0 to 14.2 percent across estimators. The IV estimate is roughly twice OLS in absolute magnitude, consistent with positive selection into CPC placement. CPC expansion over the sample period averted roughly 480 to 970 abortions per year in North and South Carolina. The aggregate pregnancy rate response is small and not statistically distinguishable from zero, pointing to substitution from abortion to birth as the dominant mechanism. Demand-side persuasion therefore produces fertility effects of the same order as supply-side access shocks, but through a different channel. Among teens, marginal births have higher low birth weight rates, and welfare concerns concentrate there (Almond et al., 2005; Black et al., 2007). Post-*Dobbs*, state funding has expanded CPC capacity while supply-side restrictions have contracted the abortion outside option.

2 Background

Crisis pregnancy centers first appeared in the late 1960s, before *Roe v. Wade*. They grew with the pro-life movement that organized in response to state-level abortion liberalization and the 1973 federal rulings (Munson, 2008). One of the earliest opened in March 1970 on Maui, founded by Robert Pearson just before Hawaii became the first state to legalize abortion. Pearson stated

his rationale plainly: “I just want to let women know that they have alternatives to abortion” (Lueras, 1973). CPCs are organizationally distinct from clinic-protest groups and focus on one-on-one counseling and material aid (Munson, 2008).¹

CPCs offer pro-life counseling, pregnancy testing, material aid (baby clothes, cribs, diapers, direct financial assistance), adoption referrals, and abstinence-focused sex education (Care Net, 2018). Most affiliate with evangelical Christian or Catholic networks such as Care Net, Heartbeat International, or the National Institute of Family and Life Advocates (NIFLA) (Swartzendruber et al., 2018b). About three-quarters offer nondiagnostic ultrasounds. Only 26 percent have a registered nurse on staff and 16 percent have a physician (Lambert et al., 2025; Desai et al., 2025; Montoya et al., 2022), so many ultrasounds occur without physician or registered-nurse involvement. CPC services thus operate through several possible channels, including counseling, material support, religious or moral appeals, abstinence education, and visual exposure to an ultrasound image. Evidence from clinical settings indicates that ultrasound viewing is associated with continuing a pregnancy primarily among the 7.4 percent of women with medium or low decision certainty (Gatter et al., 2014), suggesting that visual exposure matters near the decision margin rather than overriding strong preferences.

CPCs are small and mostly volunteer-run. IRS Form 990 filings for 68 CPC organizations operating 146 locations in NC and SC show median revenue of \$155,000 per organization and combined annual revenue of about \$15.4 million during 2015–2017. Over 60 percent report zero employee compensation. Imputing volunteer labor at the QCEW average wage for social service occupations puts the combined service value near \$23 million (Online Appendix §OA.4.1). Public funding of about \$1 million annually during the study period accounts for 4–7 percent of total CPC resources, and each public dollar leverages \$15–23 in private donations and volunteer labor (Online Appendix Tables OA.8–OA.9).

Most CPCs are not licensed medical facilities and therefore fall outside parts of the regulatory framework that governs clinical providers. The Supreme Court’s decision in *NIFLA v. Becerra* (Supreme Court of the United States, 2018) struck down California’s disclosure requirement on First Amendment grounds, limiting states’ ability to mandate non-medical-status disclo-

¹CPCs are also distinct from the strand of activism that has used violence against abortion providers, with measurable effects on abortion services (Jacobson and Royer, 2011).

asures. Audit and content-analysis studies document inaccurate or misleading health information in some CPC settings. A mystery client study of 86 New York CPCs found that 76 percent of visits included inaccurate or inflammatory statements (Frasik et al., 2023). Warren et al. (2025) document systematic false health claims in Care Net and Heartbeat International training materials.² Recent evidence from Swedish administrative records finds no causal effect of abortion on mental health among young women (Janys and Siflinger, 2024), addressing a category of claim that has appeared in CPC counseling. Over 85 percent of women in a nationally representative survey could not correctly identify at least one CPC website as non-medical (Swartz et al., 2021). Among visitors, 58.5 percent of emerging adults in Georgia did not know that the center opposes abortion (Swartzendruber et al., 2021). The data used here do not directly observe what happens in a one-on-one counseling session, but given the documented information environment, the welfare implications of CPC-induced behavior change depend on assumptions about information quality, a point developed in §4 and §6.2.

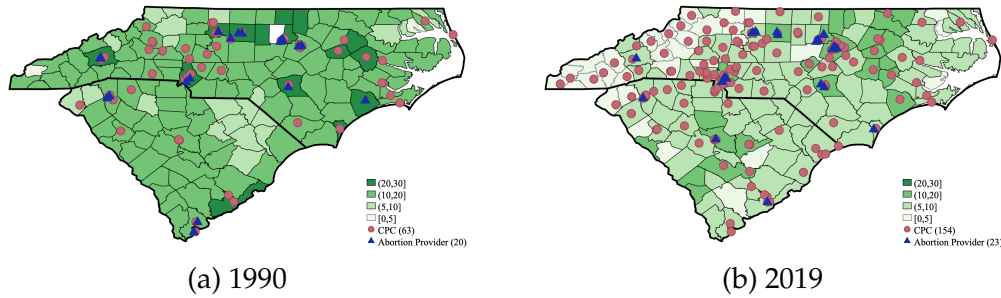
The existing empirical literature has described CPC location choice and attendance patterns descriptively (Yuengert and Fetzer, 2010; McVeigh et al., 2017). CPC attendance is not rare. Multi-state representative estimates find that 12–20 percent of women with a pregnancy history have visited a CPC (Yang et al., 2025; Rice et al., 2021), with attendance higher among Black women, lower-income women, and women without college degrees (Rice et al., 2021). Cartwright et al. (2021) find that 13.1 percent of women searching for abortion services online visited a CPC during their pregnancy, and that CPC visitors were 21 percentage points less likely to have had an abortion. That association is difficult to interpret causally because CPC attendance is selected. Women who visit CPCs may already be ambivalent about abortion, or CPCs may locate where abortion demand is high for unobserved reasons. Descriptive evidence shows high attendance and low awareness of mission but cannot identify the causal effect of a CPC visit on abortion or fertility decisions.

²The pattern has been documented for several decades. A congressional report on federally funded CPCs documented inaccurate claims about abortion risks (Waxman, 2006), public health studies found similar patterns on CPC websites (Lin and Dailard, 2002; Rosen, 2012; Swartzendruber et al., 2018a), and earlier yellow-pages advertising used ambiguous phrases such as “Considering Abortion?” and “Abortion Information” (Munson, 2008).

2.1 The CPC Network in North and South Carolina

The CPC network has expanded steadily since the early 1970s. In my sample, the number of unique CPC locations in North and South Carolina grew from 60 in 1990 to 146 by 2019.³ Growth was gradual rather than concentrated around particular policy events, consistent with decentralized decisions by individual organizations responding to local conditions. By 2019, the typical county resident lived within 8 miles of a CPC, down from over 20 miles in 1990. CPCs and abortion providers overlap geographically: by 1990, all but one county with an abortion provider already had a nearby CPC, and subsequent CPC expansion targeted underserved areas farther from existing centers (Online Appendix §OA.4). This co-location pattern is national: two-thirds of abortion facilities have a CPC within 3 miles, and the national CPC-to-clinic ratio is 3.4 to 1 (Swartzendruber et al., 2024).⁴

Figure I: CPCs, Abortion Providers, and Abortion Rates: NC & SC



Notes: 146 counties in North Carolina and South Carolina. Panels show 1990 (left) and 2019 (right). Circles denote operating crisis pregnancy centers (CPCs); triangles denote operating abortion providers. Both location types are matched to the corresponding county at their reported street addresses. County shading reports the resident abortion rate per 1,000 women aged 10–44 in 5-unit bins (legend in the figure). Counts: CPC locations grew from 60 in 1990 to 146 by 2019; the number of abortion providers was 36 in 1990 and 35 in 2019, with substantial within-state churn. CPC data from the CPC360 directory cross-referenced with state non-profit filings; abortion-provider data from the NC and SC vital statistics offices supplemented with Guttmacher Institute counts.

³Table I reports population-weighted county-level means of 1.12 CPCs in 1990 and 2.79 in 2019, which are higher than the unweighted means (0.40 and 0.97) because larger counties host more centers. The treatment variable counts all CPCs within a county's catchment area, including centers near county borders that serve multiple counties, so the sum of county-level counts exceeds the number of unique locations statewide.

⁴The competition extends to online channels, with the national network Care Net reportedly entering into bidding wars with abortion providers over sponsored-link placements on search engines (Gibbs, 2007), positioning CPCs to reach women already searching for abortion services.

2.2 Policy Environment

Throughout the study period (1990–2019), abortion was legal in both states, though subject to parental-involvement and waiting-period requirements. North Carolina required parental consent for minors beginning in 1995 and enacted a 72-hour waiting period in 2015. South Carolina required parental consent beginning in 1990 and introduced a 24-hour waiting period in 1995. Joyce et al. (2020) document that parental involvement laws enacted before the mid-1990s reduced minor abortions by 15–20 percent but that post-mid-1990s laws have no detectable effect, consistent with the modest implied confound for the NC and SC laws in this sample period (see also Bitler and Zavodny, 2002, on an earlier period). Neither state enacted targeted regulation of abortion providers (TRAP) laws severe enough to close facilities during the sample period. The FDA approval of mifepristone in 2000 introduced medication abortion as an alternative to surgical procedures, potentially lowering some clinical and nonpecuniary costs of abortion access in the second half of the sample period. I control for state-level parental involvement and TRAP policy changes directly, and the CPC coefficient is unchanged (Online Appendix Table [OA.11](#)). The relative stability of the provider landscape helps isolate CPCs as a demand-side influence in a setting where women retained legal access to abortion services.

The economics literature on mandatory counseling and waiting periods provides an important foil. These policies reduce abortion rates primarily through access costs (extra trips, travel, delay) rather than through the counseling content itself: Joyce et al. (2009) conclude that mandated counseling has little independent effect when the in-person visit requirement is not binding.⁵ CPCs differ institutionally. They do not raise the travel or delay costs of obtaining an abortion in the way waiting periods or clinic closures do; any effect of CPC exposure is therefore more naturally interpreted as operating through demand-side channels, including information, persuasion, moral suasion, and material support.

The regulatory environments around abortion provision and CPC operation are structurally asymmetric. Abortion providers face state licensing, mandatory counseling and waiting period requirements, gestational limits, parental involvement laws, and (after *Dobbs*) outright bans

⁵Lindo and Pineda-Torres (2021) similarly find that Tennessee’s mandatory waiting period increased second-trimester abortions through delay, not through persuasion. Myers (2017) frames waiting periods as generating variation in access costs rather than information.

in 18 states. CPCs generally do not face the abortion-specific licensing, gestational, and procedural regulations that apply to clinical providers, and most are not licensed medical facilities. This asymmetry matters for external validity. Post-*Dobbs* restrictions directly constrain clinical providers, while CPCs can continue expanding through counseling, referrals, material aid, and public funding programs.

CPC funding has a separate policy dimension. During the study period, NC and SC CPCs received \$1 million annually in state funding, constituting 4–7 percent of total resources (Online Appendix §OA.4). At the federal level, CPCs in 49 states received \$140 million between 2017 and 2023 through recurring programs, predominantly Temporary Assistance for Needy Families (TANF) block grants (\$103 million), with smaller allocations through Title V sexual risk avoidance education and the Teen Pregnancy Prevention Program (Rabinovitz et al., 2024). TANF’s statutory purposes include reducing out-of-wedlock pregnancies and promoting two-parent families, which provides one statutory rationale under which CPC funding has been allocated. An additional \$289 million reached CPCs through CARES Act pandemic relief. Following *Dobbs*, state legislatures directed \$489 million to CPCs nationally in two years (CBS News, 2024), and CPC growth has been 1.8 times faster in states with dedicated grant programs (Lambert et al., 2025). The scale of post-*Dobbs* public investment means that the cost-effectiveness questions in §6.2 apply to substantially larger treatment doses than those observed in the study period.

3 Data

I focus on North and South Carolina for three reasons. The local CPC-to-abortion-provider ratio is 6.7-to-1, more than double the national 3-to-1, generating substantial cross-county variation in CPC exposure for identification. Both states had legal abortion access combined with parental consent and waiting periods, a regulatory environment shared by most states in the pre-*Dobbs* South. The joint sample keeps most cross-state abortion travel internal, since NC data track residents across all states and SC data cover SC, NC, and Georgia. The panel begins in 1990, the earliest year both states report fertility counts at the county-by-age level. Summary statistics are in Table I.

3.1 CPC and Abortion Provider Data

I construct the CPC registry from IRS Form 990 filings, organizational databases, and provider directories, cross-referencing sources to obtain precise addresses and dates of operation. The registry contains 288 address-level location records linked to 138 CPC organizations, where each record is a unique organization–county–address combination: a center that changes address, or relocates across counties, generates a separate record. The treatment variable—county CPCs per 10,000 women—is built at the county-year level from these records, so a cross-county relocation shifts treatment from the origin to the destination county, while a within-county address change in the year of relocation is de-duplicated to avoid double-counting.⁶ The resulting operating stock grows from 60 locations in 1990 to 146 by 2019. The data also include 43 abortion provider locations, combining Title X grant recipient records, state licensing records, and provider lists from the National Abortion Federation and Planned Parenthood, verified against Myers’ Abortion Facility Database (Myers and Ladd, 2020).

The main treatment variable is the lagged number of CPCs per 10,000 women ages 10–44 in a county, following the per-capita scaling in Lindo et al. (2020). The treatment is lagged one year to avoid simultaneity and to reflect the institutional reality that CPC effects operate through established community presence rather than contemporaneous opening. Over the sample period, mean CPC presence doubled (from 1.1 to 2.8 per county) while mean distance to the nearest CPC fell from 20 miles to 8 miles (Table I).

3.2 Fertility Outcomes

The primary outcomes are the log abortion rate and log birth rate per 1,000 women in a county and year. Counts are obtained from administrative vital records provided by the NC and SC vital statistics offices, reported by county of residence and age group (10–19, 20–24, 25–29, 30–34, 35–44). I also construct a pregnancy rate per 1,000 women, defined as the sum of abortion counts, birth counts, and fetal death counts in each county-year cell.⁷

⁶For the funding and service-value calculations (Online Appendix §OA.4.1), I use the subset of 68 organizations with IRS Form 990 filings during 2015–2017.

⁷Fetal deaths are a small share of pregnancies (roughly 1 percent of the sum) and do not meaningfully affect the results. Miscarriages are not fully reported and are excluded, so the pregnancy rate is an undercount of total conceptions; 13 percent of clinically recognized pregnancies end in miscarriage (Andersen et al., 2000).

Two features of the data require attention. NC resident abortion data include abortions performed in any state, while SC data are limited to abortions performed in SC, NC, or Georgia, which attenuates estimated CPC effects in South Carolina.⁸ A small share of cells also report zero abortions.⁹ County-level abortion rates declined from 16.8 per 1,000 women in 1990 to 8.6 in 2019; teen rates fell from 19.0 to 2.9.

For the marital decomposition, I construct county-age-year married and unmarried female population estimates by interpolating marriage rates between Census/American Community Survey (ACS) benchmark years and applying the interpolated shares to NBER intercensal population counts.¹⁰ These marital-specific populations serve as denominators for the Poisson exposure variable, so that coefficients measure changes per married or unmarried woman rather than per woman overall.

The empirical specifications control for time-varying county characteristics that enter both the hazard model (§4.2.1) and the second-stage equation, including distance to the nearest abortion provider, demographics, economic conditions, political ideology, and religious composition (Table I lists the full set). All controls are lagged one year. County and year fixed effects are included in all specifications.

⁸South Carolina captures abortions performed in the state and in both of its bordering states (North Carolina and Georgia), so only procedures obtained farther afield are missed—a small share given the legal, in-region abortion access that prevailed throughout the study period.

⁹At the county-year level, fewer than 1 percent of abortion observations are zero; these are dropped from the log specification. The Poisson control function retains all observations. Zero-count shares rise substantially in demographic subgroup cells (up to 39 percent for marital-age cells), motivating the Poisson estimator for subgroup decompositions (Online Appendix §OA.2).

¹⁰Benchmark years are 1990 STF3, 2000 SF3, and ACS 5-year estimates for 2009, 2013, 2018, and 2019.

Table I: Summary Statistics

	1990–2019		1990		2019	
	Mean	SD	Mean	SD	Mean	SD
<i>Abortion rate (per 1,000 women)</i>						
Total	11.07	(4.90)	16.81	(5.95)	8.63	(3.14)
Age 10–19	7.86	(5.35)	19.02	(6.56)	2.91	(0.99)
<i>Birth rate (per 1,000 women)</i>						
Total	45.87	(15.03)	49.45	(13.39)	42.02	(13.44)
Age 10–19	22.06	(10.28)	34.09	(8.86)	9.73	(3.63)
<i>Pregnancy rate (per 1,000 women)^b</i>						
Total	57.4	(17.6)	66.7	(16.4)	50.9	(15.1)
<i>Crisis Pregnancy Centers</i>						
No. of CPCs	2.15	(2.46)	1.12	(1.39)	2.79	(2.80)
CPCs per 10,000 women	0.35	(0.39)	0.21	(0.30)	0.40	(0.42)
Distance to nearest CPC (miles)	11.61	(14.30)	20.33	(20.84)	7.78	(9.52)
Annual CPC revenue (\$000s) ^a	463	(1,013)	302	(711)	797	(1,313)
<i>County Characteristics</i>						
Female population age 10–44	67,406	(69,966)	48,246	(42,286)	88,599	(92,885)
Non-white share	0.32	(0.15)	0.29	(0.15)	0.34	(0.15)
Share age 10–19	0.27	(0.03)	0.26	(0.03)	0.28	(0.02)
Nearest abortion clinic (miles)	27.33	(25.90)	36.60	(28.31)	30.73	(26.33)
Unemployment rate	5.95	(2.66)	4.36	(1.54)	3.56	(0.79)
U.S. House GOP vote share	0.53	(0.20)	0.45	(0.24)	0.52	(0.18)
Protestant share	0.54	(0.12)	0.60	(0.12)	0.51	(0.11)
Catholic share	0.04	(0.03)	0.02	(0.02)	0.04	(0.03)
No. of county-year obs.	4,380		146		146	

Notes: Population-weighted county-level means; standard deviations in parentheses. 146 counties observed annually, 1990–2019.

Sources: Abortion counts from NC and SC vital statistics offices (county of residence). Birth and fetal death counts from NBER Vital Statistics Natality and Fetal Death files. CPC locations from author-constructed registry cross-referenced with IRS filings, Care Net and Heartbeat International directories, and state licensing records. CPC revenue from IRS Form 990 filings via NCCS Core Files; ^aThe 1990 column for CPC revenue reports 1997 values (earliest year with non-trivial IRS filing data for matched CPCs). Abortion provider locations from Title X grant records, state licensing records, National Abortion Federation and Planned Parenthood directories, and Myers' Abortion Facility Database (Myers and Ladd, 2020). Population and demographics from NBER intercensal population estimates (SEER). Unemployment rate from Bureau of Labor Statistics, Local Area Unemployment Statistics. U.S. House GOP vote share from CQ Voting and Elections Collection. Religious adherence from Association of Religion Data Archives, Religious Congregations and Membership Study (1990, 2000, 2010), linearly interpolated.

4 Empirical Framework

CPCs affect the decision of a woman who is on the margin, uncertain whether to have a child or terminate the pregnancy. This is a potentially large share of women, because 45 percent of all pregnancies are reportedly mistimed or unwanted (Finer and Zolna, 2016). The second group of women who are expected to be more likely to visit a CPC are those facing substantial barriers, for example a great travel distance, to access abortion services from a medical provider. They may be looking for any available support or mistake a CPC for an organization that provides abortion services.

Consider a pregnant woman deciding whether to carry to term or abort. She compares the net benefit of birth, B_i (relationship status, desired fertility, income, social support), against the net benefit of abortion, A_i (access costs, stigma, foregone education, labor market participation). She chooses an abortion if $S_i \equiv A_i - B_i > 0$. The surplus S_i is the woman’s private valuation of abortion over birth given her information at the time of the decision.

A CPC shifts this margin. Through counseling, ultrasound exposure, material support, and community referrals, a CPC reduces the woman’s perceived surplus by some amount $\delta_{\text{CPC}} > 0$. A woman is redirected from abortion to birth when her pre-CPC surplus was positive but small, $S_i \in (0, \delta_{\text{CPC}})$. This is the *substitution* channel. Abortions fall one-for-one with births rising, and the pregnancy rate is unchanged.

Whether this redirection improves or reduces welfare depends on the quality of the information driving it.¹¹ If the woman’s pre-CPC beliefs overestimated the cost of carrying to term, because she lacked information about available support, adoption procedures, or her own eligibility for assistance, counseling that corrects the overestimate brings her decision closer to her informed preferences. If CPCs present misinformation, the client’s prior S_i is distorted.¹² The county-level data here cannot identify which interpretation dominates. The welfare analysis in §6.2 reports the implications under each polar case, and the birth-quality evidence for teenagers shifts plausibility weight toward the distortion case in that subgroup.

A second channel is *deterrence*. If CPCs shift upstream behavior through abstinence-focused education or community influence, some women who would have become pregnant do not. Abortions, births, and pregnancies all fall.¹³ The framework treats deterrence as a residual: it is identified by a reduction in pregnancies not accompanied by a corresponding rise in births. The empirical implementation tests for it via the pregnancy-rate response (§5) and by age (§5.1).

Table II summarizes the testable predictions. Both channels reduce abortions, but they

¹¹The estimated δ_{CPC} in §5 is the net effect of CPC counseling relative to the woman’s counterfactual decision process, including any information she would have obtained from a doctor, family member, or independent research, not relative to no information at all. Dahl and DellaVigna (2009) make the analogous point for media exposure, where field estimates capture the difference between the treatment and the foregone alternative activity.

¹²See §2 for documentation of these distortion channels.

¹³The evaluation literature largely finds no effect of abstinence-only education on teen sexual activity or pregnancy: a Mathematica RCT (Trenholm et al., 2008), an NSFG cross-section (Kohler et al., 2008), and a state-level quasi-experimental study (Carr and Packham, 2017). Cannonier (2012) finds birth-rate reductions limited to white 15–17-year-olds.

Table II: Predicted Effects by Channel

Channel	Δ Abortions	Δ Births	Δ Pregnancies
Substitution	< 0	> 0	≈ 0
Deterrence	< 0	< 0	< 0
Total effect	< 0	indeterminate	≤ 0

Notes: Predicted signs of each mechanism’s effect on county abortion, birth, and pregnancy rates (§4). The pregnancy-rate response discriminates between the channels: a null rules out pure deterrence, a negative effect rules out pure substitution (§5).

differ on births and pregnancies. The pregnancy rate is the cleanest discriminator: a null effect rules out pure deterrence; a negative effect rules out pure substitution. The sign of the net birth effect is empirically determined by the relative magnitudes of substitution and deterrence.

The mix of substitution and deterrence may vary with age, since the two channels operate on different states of the woman. Substitution requires that the woman be already pregnant when she contacts a CPC. Deterrence operates on women who have not yet conceived. Whether a given CPC interaction is with a pregnant or pre-pregnant woman depends on the referral source, which varies across age. CPC services include campus outreach and community programming that reach pre-pregnant young women, alongside in-person counseling that women self-refer to during a pregnancy. The framework does not place a functional form on this variation.

4.1 Estimation Strategy

The goal is to estimate how CPC presence affects fertility outcomes. The fundamental obstacle is that CPC placement is endogenous. CPCs locate strategically.¹⁴ The founders who decide where to open a center observe local demand for abortion alternatives and unmet need that an outside researcher cannot observe. The same unobserved factors that drive CPC entry may also drive fertility outcomes. Estimating the effect of CPC presence by OLS conflates the causal effect with this selection.

The endogeneity has a second, less familiar dimension. CPC entry is a dynamic, path-dependent process. The probability that a county receives its k th CPC depends on the full history of prior placements: each opening changes the local CPC stock, the distance to the nearest center,

¹⁴Munson (2008) notes that CPCs are sometimes sited next door to or directly across the street from abortion clinics, an extreme form of the location endogeneity the simulated instrument addresses.

and the remaining unserved population, all of which feed back into the probability of the next opening. Over thirty years, this compounding produces a nonlinear function of past observables and organizational timing shocks that no standard linear specification can capture.

I address both problems with a single instrument constructed from a hazard model of CPC entry. The approach has three steps. First, I estimate a discrete-time logit for CPC openings as a function of lagged county characteristics. Second, I forward-simulate counterfactual CPC expansion paths, averaging across draws to eliminate organizational timing shocks and isolate the component of CPC presence predicted by predetermined fundamentals. Third, I use this simulated measure as an instrument for actual CPC presence in a standard 2SLS framework. The structural equation estimated separately for abortion rates, birth rates, and pregnancy rates is:

$$y_{c,t} = \theta_0 + \theta_1 \text{CPC}_{c,t-1} + \theta_2' X_{c,t-1} + \mu_c + \lambda_t + u_{c,t}, \quad (1)$$

where $y_{c,t}$ is the natural log of the outcome in county c and year t ; $\text{CPC}_{c,t-1}$ is the lagged CPC exposure measure; $X_{c,t-1}$ is a vector of lagged time-varying county characteristics; and μ_c and λ_t denote county and year fixed effects. Estimating θ_1 by OLS requires strict exogeneity: $E[u_{c,t} \mid \{\text{CPC}_{c,\tau-1}, X_{c,\tau-1}\}_{\tau=1}^T] = 0$. This condition fails under strategic placement, and the instrument is designed to address it.

4.2 Instrument Construction

4.2.1 Hazard Model of CPC Entry

The first step is to model where and when CPCs open. I estimate the probability that a county receives a new CPC in each year as a function of its current characteristics, using a discrete-time logit:

$$p_{c,t} = P(\text{NewCPC}_{c,t} = 1 \mid W_{c,t-1}) = \frac{\exp(\beta' W_{c,t-1})}{1 + \exp(\beta' W_{c,t-1})} \quad (2)$$

Counties can receive multiple CPCs; the model allows repeated events by conditioning on the accumulated stock $\text{CPC}_{c,t-1}$.¹⁵ The conditioning set $W_{c,t-1}$ includes lagged CPC stock, distance to

¹⁵This extends the single-spell hazard to allow multiple openings following Mroz (2012). Details are in Appendix A.1.

nearest CPC, lagged abortion rates, abortion provider presence, demographics, religious composition, political ideology, unemployment, county fixed effects, and a cubic polynomial time trend. The hazard specification builds on the empirical non-profit location literature (Wolch and Geiger, 1983; Wolpert, 1993; Bielefeld and Murdoch, 2004), which models facility placement as a function of local needs, resources, and existing supply. Full specification details and hazard model estimates appear in Appendix A.

4.2.2 Forward Simulation

The instrument answers a counterfactual question: what would CPC expansion have looked like in each county if organizational timing were random but location fundamentals were held fixed? Starting from observed 1990 conditions, I use the estimated hazard to predict opening probabilities in each county-year, then draw independent random shocks to determine whether a CPC opens. Each opening updates the county’s CPC stock and distance measures, which feed into the next year’s predicted probability (Appendix Table A1 details the variable treatment protocol). Repeating this forward simulation $R = 1,000$ times and averaging across draws yields:

$$IV_{c,t} = \frac{1}{R} \sum_{r=1}^R CPC_{c,t-1}^{(r)} \xrightarrow{p} \mathbb{E}[CPC_{c,t-1} \mid W_{c,t-1}] \quad (3)$$

where $W_{c,t-1}$ denotes the full set of lagged county characteristics entering the hazard model. Path dependence makes closed-form computation of this conditional expectation intractable, and the forward simulation is a Monte Carlo approximation (McFadden, 1989).

The key point is what the averaging eliminates and what it preserves. The actual timing of CPC openings depends on organizational shocks: funding cycles, real estate availability, volunteer recruitment, denominational approval. These shocks are plausibly idiosyncratic and uncorrelated with contemporaneous fertility disturbances. The simulated shocks are computer-generated random numbers standing in for these unobserved timing factors. Averaging 1,000 draws eliminates them by the law of large numbers, leaving a deterministic function of $W_{c,t-1}$. No actual organizational shocks enter the instrument.

What remains after averaging is the nonlinear structure that the hazard model imposes on predetermined county characteristics. If a CPC opens in 2003, it changes the probability of

another opening in 2010 through several channels: the local CPC stock rises, the distance to the nearest center falls, and the unserved population shrinks. The logit link, cumulative stock, recursive distance updating, and compounding over thirty years combine these effects nonlinearly. The resulting cross-county variation in predicted CPC trajectories cannot be replicated by any linear function of the underlying covariates, and it is this residual nonlinear variation that the first stage uses. Regressing $IV_{c,t}$ on predetermined observables confirms this: county and year fixed effects capture 77 percent of the instrument’s variation; adding the linear hazard covariates raises R^2 to 0.80; quadratic terms and interactions raise it only to 0.82. First-stage power comes from a quantitatively modest but statistically potent nonlinear component (first-stage $F = 196.1$; see §E.2).¹⁶

Methodological contribution. The instrument combines elements of two methodological traditions. The first is simulated instruments, pioneered by Currie and Gruber (1996), who apply each state’s Medicaid eligibility rules to a fixed national reference sample, isolating variation from legislative changes. Their exclusion restriction is that state rules are exogenous to health outcomes. This paper adapts the simulation logic to a setting without exogenous policy rules: instead of applying deterministic rules to a reference population, I simulate counterfactual expansion paths from a hazard model of each county’s own predetermined characteristics. The exclusion restriction is correspondingly different; it holds if lagged observables are predetermined with respect to current fertility shocks (equation 4).

The second tradition is the joint modeling of endogenous program placement and outcomes in Angeles et al. (1998), who estimate a discrete-time hazard for family planning facility openings simultaneously with fertility outcomes. This paper uses the hazard model only to construct the instrument, separating the placement prediction from the outcome estimation. The combination is a hazard-based instrument that inherits exogeneity from predetermination rather than from exogenous policy rules or joint estimation. The approach is portable to any setting where facility or program placement is path-dependent and there is no natural experiment to exploit.

The panel structure provides additional identification leverage. With 10 time-varying co-

¹⁶The IV estimate is stable (-0.109 to -0.177) when different categories of time-varying controls are dropped from the second stage (§E.2), providing an informal over-identification check.

variates and 30 years, predetermination generates 290 moment conditions (Bhargava, 1991; Liu et al., 2010). Exploiting these individually would create a many-weak-instruments problem. The forward simulation aggregates them into a single strong instrument through the hazard model, imposing an economically motivated functional form that yields a first-stage F of 196.1.

4.3 Identification

The identification argument rests on a single assumption about the information structure of the data. The exclusion restriction follows from it mechanically rather than requiring a separate institutional argument.

Predetermination Assumption. The structural error $u_{c,t}$ has mean zero conditional on lagged observables, county and year fixed effects:

$$\mathbb{E}[u_{c,t} \mid W_{c,t-1}, \mu_c, \lambda_t] = 0. \quad (4)$$

This assumption has a transparent economic interpretation. County fixed effects absorb all time-invariant differences across counties, including geographic, cultural, and institutional factors that determine both baseline CPC presence and baseline fertility patterns. Year fixed effects absorb aggregate trends affecting all counties simultaneously, including national policy changes, macroeconomic conditions, and secular shifts in fertility norms. Lagged abortion rates and demographic covariates in $W_{c,t-1}$ absorb persistent county-specific demand dynamics that evolve predictably from one period to the next. What remains in $u_{c,t}$ are period-specific, county-specific fertility shocks that prior observables cannot forecast: an unexpected local economic downturn, an unanticipated change in clinic access, or an idiosyncratic shift in contraceptive behavior.

The assumption would fail if a county-level shock simultaneously increased unwanted pregnancies and triggered a CPC opening in the same year. The one-year lag in $W_{c,t-1}$ provides the first line of defense; the economic basis for the assumption is that CPC openings are driven by organizational processes—fundraising cycles, real estate negotiations, volunteer recruitment, denominational approval—that unfold over months or years in response to community characteristics. These organizational frictions determine the timing of CPC entry but operate on different timescales and through different channels than the contemporaneous shocks that drive annual

county-level fertility fluctuations.

Predetermination is weaker than strict exogeneity: it conditions only on $W_{c,t-1}$ and earlier, so it permits current fertility shocks $u_{c,t}$ to feed back into future observables $W_{c,t+1}$. This matters because county-level demographics and socioeconomic conditions plausibly respond to fertility shocks with a lag. Standard two-way fixed effects estimation under parallel trends implicitly requires strict exogeneity of treatment status—ruling out such feedback—which is a stronger requirement in settings where treatment timing may respond to lagged outcomes. The IV requires only that lagged observables are uncorrelated with *current* shocks, making it the appropriate primary strategy.

Predetermination delivers the exclusion restriction without requiring any separate argument. Since the instrument converges to $\mathbb{E}[\text{CPC}_{c,t-1} \mid W_{c,t-1}]$ and is therefore measurable with respect to $W_{c,t-1}$, the law of iterated expectations implies:

$$\mathbb{E}[u_{c,t} \cdot \text{IV}_{c,t} \mid \mu_c, \lambda_t] = \mathbb{E}\left[\text{IV}_{c,t} \cdot \underbrace{\mathbb{E}[u_{c,t} \mid W_{c,t-1}, \mu_c, \lambda_t]}_{=0}\right] = 0. \quad (5)$$

More broadly, instruments derived from predetermined covariates inherit their exogeneity from the information structure of the data rather than from institutional arguments about channels.¹⁷

The hazard model determines instrument strength and complier weighting, not the exclusion restriction. Misspecifying the hazard weakens the first stage but does not introduce bias as long as predetermination holds. The Monte Carlo validation (§4.5) confirms this: the hazard is deliberately misspecified in all cells, yet the IV recovers the true parameter with bias below 0.07.

4.3.1 Threats to Identification

The instrument is valid if no county-level fertility shock in year t can be forecast from lagged county observables (equation 4). This requires that CPC entry timing is not driven by the same factors that drive contemporaneous abortion demand, and that the lagged covariates enter-

¹⁷This principle connects the identification strategy to the formula instruments framework of Borusyak et al. (2025) and the quasi-experimental shift-share literature (Borusyak et al., 2022; Goldsmith-Pinkham et al., 2020), where Goldsmith-Pinkham et al. (2020) require that initial industry shares are exogenous to local labor demand shocks. The instrument here is not a shift-share instrument—there are no “shares” and “shocks” to decompose—but the identification logic is parallel.

ing the hazard model are not endogenous to current outcomes through unmodeled persistence. The first three threats below address the assumption directly: anticipation, correlated provider access changes, and persistent unobservables. The remaining two diagnose whether the assumption holds in practice, by testing residual orthogonality and by checking that instrument strength does not rely on the time-trend specification. Full results are in Online Appendix [OA.5](#).

Anticipatory responses. If organizations or individuals anticipate CPC openings and adjust behavior in advance, pre-treatment fertility trends would differ between treated and control counties. Event study estimates (Figure [II](#)) show flat pre-trends for abortion, birth, and pregnancy rates, with joint F -tests failing to reject the null ($F = 1.11$, $p = 0.359$ for abortion; $F = 0.88$, $p = 0.547$ for birth; $F = 1.00$, $p = 0.446$ for pregnancy). The joint tests cannot reject flat pre-trends for any outcome, though two of the nine individual pregnancy leads (event times -5 and -4) are statistically significant; the mechanism evidence accordingly rests primarily on the continuous-intensity IV estimates rather than the binary event study ([§5](#)).

Correlated provider access changes. If CPC entry coincides with abortion clinic closures, the instrument would capture supply-side effects rather than demand-side CPC persuasion. Restricting to counties always more than 30 miles from the nearest abortion provider yields an estimate of similar magnitude (-0.149 , versus -0.142 at baseline) with a strong first stage ($F = 173.9$; Online Appendix Table [OA.11](#)).

Persistent unobservables. If twice-lagged unobservables affect both current CPC presence (through the hazard model’s use of lagged variables) and current fertility outcomes, predetermination would fail through a backdoor channel. Controlling for twice-lagged abortion rates attenuates the estimate but it remains significant, and the first stage remains strong ($F = 135.9$; Online Appendix Table [OA.10](#)). The attenuation is consistent with some persistence in the outcome process, not with a violation of the exclusion restriction. This specification provides a lower bound on the causal effect under partial departures from predetermination; the range between the baseline and lagged-outcome-controlled estimates bounds the effect under varying degrees of outcome persistence ([§5.2](#)).

Serial correlation diagnostic. The direct predetermination test is whether lagged observables predict the 2SLS residual. Panel C of Appendix Table [A5](#) shows they do not ($F < 0.01$, $p > 0.99$, within $R^2 = 0.0000$), indicating that the instrument is orthogonal to the structural error conditional

on the control set. The 2SLS residuals exhibit serial correlation ($AR(1) = 0.41$, $AR(2) = 0.33$; Panel B), reflecting the natural persistence of county-level abortion rates in a levels specification with county fixed effects. Progressive lagged-outcome controls absorb this persistence, and the 2SLS coefficient stabilizes at the Nickell (1981) bias bound (Panel A), consistent with outcome dynamics rather than confounding.

Functional form dependence. If the instrument’s predictive power comes entirely from the polynomial time trends in the hazard model (which could mechanically correlate with outcome trends), the identification would be fragile. Varying the polynomial order from linear through sextic leaves the estimate virtually unchanged (Online Appendix Table [OA.13](#)), showing that the cross-sectional variation in county characteristics, not the time-series specification, drives the instrument.

4.4 Estimation

First Stage:

$$CPC_{c,t-1} = \pi_0 + \pi_1 IV_{c,t} + \pi_2' X_{c,t-1} + \mu_c + \lambda_t + v_{c,t} \quad (6)$$

Second Stage:

$$y_{c,t} = \alpha + \gamma \widehat{CPC}_{c,t-1} + \phi' X_{c,t-1} + \mu_c + \lambda_t + u_{c,t} \quad (7)$$

where $y_{c,t}$ denotes the log fertility outcome (abortion rate or birth rate). The coefficient γ identifies θ_1 from equation (1). Because the instrument is a function of predetermined variables, 2SLS standard errors remain valid regardless of estimation error in the hazard model (Amemiya, 1974; Murphy and Topel, 1985).¹⁸ Standard errors are clustered at the county level throughout. Observations are weighted by the square root of female population for variance stabilization.

The second-stage controls match the hazard model covariates (all lagged one year), with two differences: lagged abortion rates enter the hazard model but not the second stage (county

¹⁸See also Amemiya (1985, Chapter 8). Any function of exogenous variables is a valid instrument, so replacing one such function with a nearby estimated function does not affect the asymptotic distribution. Cluster-bootstrapped standard errors (resampling counties, re-estimating the hazard, and re-simulating within each draw) are similar across all specifications.

fixed effects absorb the cross-sectional variation), and the hazard model uses polynomial time trends while the second stage uses year fixed effects (§4.2.1).

In thin cells (marital-age decompositions), zero counts make log unreliable, so subgroup decompositions use a Poisson control function (CF) with subgroup-specific population as the exposure variable.¹⁹ The 2SLS log-rate specification is the primary estimator because it imposes weaker functional form assumptions and permits standard weak-instrument diagnostics. The Poisson CF serves as a complement: it accommodates zero counts in thin demographic cells where log is undefined, but requires correct specification of the linear first stage for the control function to fully absorb endogeneity. The two estimators bracket the credible range at $[-0.142, -0.070]$.²⁰

4.5 Validation

The instrument is constructed from an estimated model, so it is natural to ask whether the procedure works when that model is wrong. Monte Carlo experiments address this question directly. The design crosses three true treatment effects ($\beta \in \{-0.10, -0.20, -0.30\}$) with three data-generating processes (DGPs) (additive confounding, nonlinear confounding in which the confounder enters as its square, and heterogeneous effects by county age composition) for nine cells, each with 500 replications. The hazard model is deliberately misspecified in all cells: the true DGP includes confounders and functional forms not available to the hazard estimator.

The IV recovers the true treatment effect with bias below 0.07 in all nine cells (Table A3, Figure A1). OLS is severely biased in the additive-confounding and heterogeneous-effects cells, with mean estimates near +1.5 when the true effect is negative, a sign reversal that illustrates how badly endogenous placement confounds the naïve estimate; in the nonlinear-confounding

¹⁹At the county-year and county-year-age level, fewer than 4 percent of abortion observations and fewer than 1 percent of birth observations are zero, so log is well-behaved. For marital-age subgroup cells, zero-count shares rise substantially (31–39 percent), motivating the Poisson estimator. The Poisson CF follows Wooldridge (2014): a linear first stage recovers residuals, which enter as a control function in the Poisson second stage with population exposure. Exposure is the marital-status-specific female population derived from Census/ACS benchmarks, so that coefficients are semi-elasticities of the married or unmarried rate.

²⁰Under correct specification, both estimators target the same local average treatment effect defined by the instrument. The Poisson CF point estimate (-0.070) lands closer to OLS (-0.072) than to 2SLS (-0.142). Two explanations are consistent with this pattern. First, the exponential conditional mean function in Poisson weights observations differently than the log-linear 2SLS, producing a different weighted average of heterogeneous effects across counties and years. Second, if the linear first stage does not fully capture the endogenous component, the control function undercorrects for selection. Weak-instrument bias is implausible as a third explanation at first-stage $F = 196.1$. The gap cannot be decomposed without individual-level data on CPC contact and pregnancy decisions.

cells the OLS bias is smaller, while the IV remains nearly unbiased throughout. In the actual data, OLS attenuates the effect toward zero (roughly half the IV magnitude), consistent with positive selection but not sign reversal. The IV’s recovery of the true β despite hazard misspecification confirms that instrument validity depends on predetermination, not on the hazard model being correctly specified. Coverage of 95% confidence intervals is near nominal across the design. A permutation test shuffling the instrument across counties yields a first-stage F of 0.66 (median) versus 196.1 in the actual data ($p = 0.000$; Appendix Table A9 and Figure A4), confirming that instrument strength is not an artifact of overfitting county characteristics. Full details appear in Appendix D.

5 Results

Crisis pregnancy centers reduce abortion rates. The IV abortion-rate estimate is -0.142 per CPC per 10,000 women, with a first-stage F of 196.1 (Table III). The aggregate pregnancy rate response is small and statistically indistinguishable from zero, ruling out full deterrence and pointing to substitution from abortion to birth as the dominant mechanism. Event study estimates around the first CPC opening (Figure II) show flat pre-trends and post-treatment patterns consistent with these results.²¹

The OLS regression of the abortion rate on CPC presence yields a coefficient of -0.072 . This is the naïve estimate that ignores the strategic-placement problem identified in §4.3. CPCs make a strategic location decision, so the OLS coefficient mixes the negative causal effect of CPC presence with a correlation between unobserved abortion demand and CPC location. The simulated IV estimate is -0.142 , twice the OLS magnitude in absolute value, with a first-stage F of 196.1. The gap is consistent with a positive-selection story, and the CF endogeneity test rejects exogenous placement (CF residual = 0.102, $p < 0.05$).

The framework in §4 gives two candidate mechanisms. Substitution from abortion to birth

²¹Joint F -tests fail to reject the pre-trend null (abortion $p = 0.359$; birth $p = 0.547$; pregnancy $p = 0.446$). Post-treatment abortion coefficients are uniformly negative, birth coefficients positive, and pregnancy coefficients near zero. The event study uses first opening as the treatment event, following standard practice in staggered adoption settings. The IV specification (§5) uses CPC count adjusted by county population, capturing intensity of exposure that the binary event-study indicator does not. The geographic expansion of CPCs from 1990 to 2019 to more than half of all counties is shown in Figure I; the staggered adoption pattern is in Online Appendix Figure OA.2.

would leave the pregnancy rate unchanged. Deterrence of pregnancies would reduce all three rates. The pregnancy null is the cleanest discriminator. Under full deterrence the coefficient would be negative; instead the data deliver a precisely estimated zero (Panel C).²² A Wald test of the joint substitution restriction cannot reject one-for-one substitution ($p = 0.30$).²³ The pregnancy null and the Wald non-rejection together rule out full deterrence and point to substitution from abortion to birth as the dominant mechanism. The aggregate birth coefficient is positive but imprecisely estimated, consistent with substitution but not confirmatory on its own; one-for-one substitution implies only a 3 percent birth increase relative to a 14 percent abortion reduction, a small signal against cross-county variance. Partial deterrence at moderate magnitudes (30–60 percent of full) is not separately identified.

The CPC effect concentrates on the women most likely to choose abortion. Across ten age-by-marital cells, the cells with the largest CPC effects on the abortion rate are the same cells with the highest baseline abortion propensity (cross-cell covariance = 0.004; Online Appendix Table OA.3, Online Appendix §OA.1), following the targeting logic in Allcott and Taubinsky (2015). This is the pattern expected under well-targeted persuasion rather than random or anti-targeted contact. A separate diagnostic shows that CPC-induced births concentrate in cells with lower baseline low-birth-weight rates, so the women redirected from abortion to birth come from cells with healthier baseline birth outcomes. This is the empirical signature of CPCs operating on the marginal woman near the abortion margin, a pattern Section 6 picks up to interpret the modest aggregate magnitudes and the small per-redirected-woman welfare cost.

²²A 14.2% reduction in the abortion rate (mean 11.1 per 1,000) implies a pregnancy reduction of 0.028 log points under full deterrence (mean pregnancy rate 57.4). The actual point estimate of +0.014 is 3.2 standard errors from this prediction, rejecting the full-deterrence benchmark at conventional levels in this sample. Ex ante power to detect the full-deterrence effect was 58% (MDE at 80% power = 0.036 against a predicted magnitude of 0.028), so partial deterrence at moderate magnitudes (30 to 60 percent of full) would not be reliably detected by this test.

²³The restriction is $H_0 : \hat{\theta}_{\text{birth}} + \hat{\theta}_{\text{abort}} \times (\bar{r}_{\text{abort}}/\bar{r}_{\text{birth}}) = 0$. Under full substitution, a 14.2% abortion reduction maps onto a 3.4% birth increase; the actual birth coefficient of +1.8% falls within the confidence interval. The test is conservative because the covariance between the abortion and birth coefficients is set to zero (separate regressions with different dependent variables); the true variance is smaller, making the non-rejection informative.

Table III: Effect of Crisis Pregnancy Centers on Abortion, Birth, and Pregnancy Rates

	(1) OLS	(2) 2SLS	(3) Poisson-CF
<i>First Stage: CPCs per 10,000 on Simulated Instrument</i>			
Simulated CPCs per 10,000		0.823*** (0.059)	
<i>Panel A: Log Abortion Rate</i>			
CPCs per 10,000	-0.072** (0.029)	-0.142*** (0.039)	-0.070* (0.041)
CF residual (endog. test)			0.102** (0.047)
N	4,376	4,376	4,380
First-stage F		196.1	
<i>Panel B: Log Birth Rate</i>			
CPCs per 10,000	0.010 (0.008)	0.018 (0.012)	0.019 (0.017)
CF residual (endog. test)			-0.015 (0.020)
N	4,380	4,380	4,380
First-stage F		196.1	
<i>Panel C: Log Pregnancy Rate</i>			
CPCs per 10,000	0.010 (0.008)	0.014 (0.013)	0.024 (0.018)
CF residual (endog. test)			-0.014 (0.020)
N	4,380	4,380	4,380
First-stage F		196.1	
County FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Controls	Yes	Yes	Yes
$\sqrt{\text{pop}}$ weights	Yes	Yes	—

Notes: Standard errors clustered at the county level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Column (1): OLS with county and year fixed effects. Column (2): 2SLS using simulated CPC instrument. Column (3): Poisson regression with control function (Wooldridge, 2014); exposure = county female population; analytic SEs. A specification jointly instrumenting CPC count and distance to nearest CPC (both endogenous regressors instrumented) yields a larger abortion coefficient (-0.265) but with a first-stage F of 2.4 (available on request). Pregnancy rate equals abortion rate plus birth rate plus fetal deaths per 1,000 women. All specifications include county demographic controls. The first stage is re-estimated on each panel's sample ($N = 4,376$ for log abortion, 4,380 for log birth and pregnancy); the Kleibergen-Paap F is 196.05 and 196.15, respectively, and rounds to 196.1 in every panel.

5.1 Heterogeneity by Age

Table IV reports separate 2SLS specifications by age group.²⁴ Three patterns hold across the age distribution. The abortion reduction is similar at all ages. The pregnancy null holds at all five ages. The birth response varies: positive and significant at ages 25 and older, statistically null at younger ages in the log-rate specification. The abortion coefficients range from -0.130 (ages 35–44) to -0.197 (ages 25–29), all negative and significant, and a joint equality test cannot distinguish them.²⁵

At ages 25 and older, substitution operates. Birth coefficients are positive and significant at every age above 25, and the implied substitution ratios bracket unity (Online Appendix Table OA.6). Averted abortions appear as additional births in roughly the magnitudes the framework predicts. The birth and abortion estimates jointly support a substitution interpretation at these ages.²⁶

For teenagers, count-level and Poisson specifications support substitution. The count-level IV on teen birth counts yields $\hat{\beta}_{\text{births}} = 44.2$ ($t = 3.43$; Online Appendix Table OA.1), significant at the 1% level, where $\hat{\beta}$ denotes a count-level IV coefficient, distinct from the log-rate coefficient $\hat{\theta}$ in equation (1). The Poisson marital decomposition localizes the effect among unmarried teenagers (abortion -0.117 , $p < 0.05$; birth $+0.170$, $p < 0.01$); the pattern is robust to a specification that fixes the marital population denominator at its 1990 baseline (Online Appendix §OA.2.2). The log-rate specification is underpowered at this age because teen birth and abortion rates fell most steeply over the panel, so the welfare analysis in §6.2 uses the count-level specification.

²⁴Each column is a separate county-year 2SLS regression (first-stage $F > 193$). Treatment is measured as lagged CPCs per 10,000 women using the county-level total female population as the denominator across all age groups, ensuring a common treatment measure since CPC services are not age-restricted. Because the unit of observation is the county-year, coefficients measure effects on county-level rates. The age and marital decompositions below make individual-level mechanism claims (e.g., which women switch from abortion to birth) that are ecological inferences requiring additional assumptions (see King, 1997; Manski, 1995, on the limits of ecological inference).

²⁵A joint F -test fails to reject equality ($F = 0.63$, $p = 0.64$). The test is conservative because the five regressions use different samples so the covariance across coefficients is zero, maximizing within-test variance. All five coefficients remain significant at the 1% level after Holm step-down correction (Online Appendix Table OA.19).

²⁶A marital decomposition at ages 25–34 is sensitive to compositional weighting and not interpretable as behavioral redirection. The married share of women aged 10–44 fell from 0.59 in 1990 to 0.35 in 2019, reflecting the secular decline in shotgun marriage that Akerlof et al. (1996) document for the post-Roe period. An unweighted decomposition produces a pattern consistent with marriage-during-pregnancy, but a synthetic-population specification that fixes the marital share at its 1990 baseline (Online Appendix §OA.2.2) reverses it, with both marital-cell birth coefficients positive at ages 25–34 ($+0.134$ at 25–29 and $+0.125$ at 30–34 for unmarried births, $+0.098$ and $+0.096$ for married births). Online Appendix §OA.2.1 reports the full Poisson control function decomposition; Online Appendix Table OA.4 reports the marital coefficients.

At ages 20–24 the channel is genuinely ambiguous. The substantive question is what fraction of averted abortions appear as additional births. All three estimators (inverse hyperbolic sine (asinh) rate 2SLS, count-level 2SLS, Poisson CF) return a small negative and statistically insignificant birth coefficient (Online Appendix Table OA.7). The asinh-rate 95% confidence interval is $[-0.060, +0.027]$. The full one-for-one substitution prediction is $+0.042$ (given a local abortion-to-birth ratio of 0.22), which lies outside this interval. The data reject full substitution but cannot reject zero. Mapping the interval onto the substitution share $s \in [0, 1]$, where $s = 1$ is full one-for-one redirection, places the upper bound at 0.60. At this age the mechanism is partially bounded rather than fully identified.

Substitution thus operates cleanly at ages 25 and older, holds for teens under count-level specifications, and is partially bounded at 20–24. Online Appendix Figure OA.1 visualizes the age-specific coefficients.

Table IV: Age Heterogeneity in CPC Effects (2SLS)

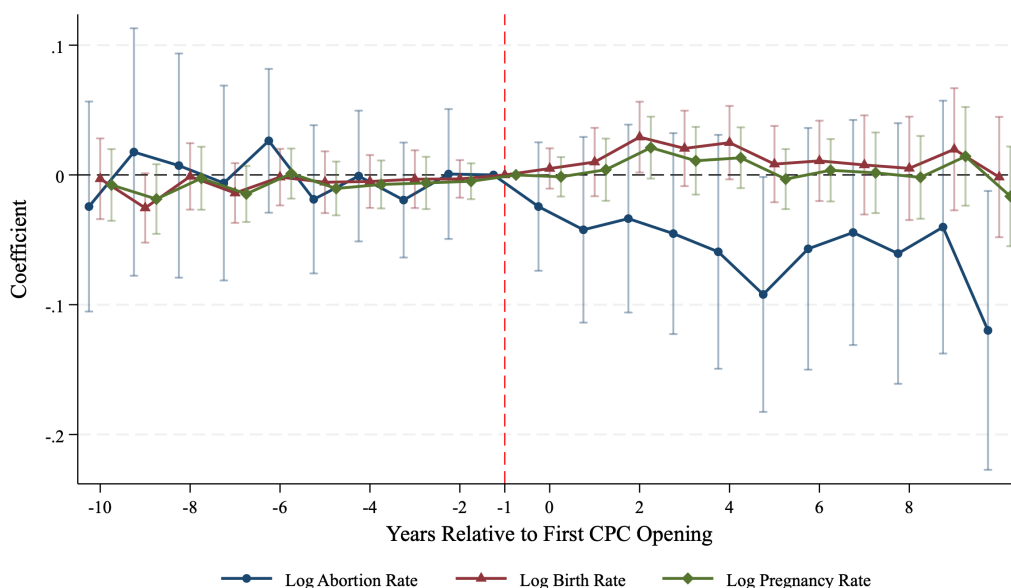
	(1) 10–19	(2) 20–24	(3) 25–29	(4) 30–34	(5) 35–44
<i>Panel A: Log Abortion Rate</i>					
CPCs per capita	-0.162*** (0.049)	-0.185*** (0.042)	-0.197*** (0.048)	-0.151*** (0.043)	-0.130*** (0.039)
N	4,380	4,380	4,380	4,380	4,380
First-stage F	193.9	206.9	207.7	207.8	193.1
<i>Panel B: Log Birth Rate</i>					
CPCs per capita	0.016 (0.029)	-0.016 (0.022)	0.032** (0.016)	0.040** (0.018)	0.050** (0.021)
N	4,380	4,380	4,380	4,380	4,380
First-stage F	193.9	206.9	207.7	207.8	193.1
<i>Panel C: Log Pregnancy Rate</i>					
CPCs per capita	0.003 (0.029)	-0.010 (0.023)	0.024 (0.016)	0.028 (0.017)	0.029 (0.021)
N	4,380	4,380	4,380	4,380	4,380
First-stage F	193.9	206.9	207.7	207.8	193.1
County FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes	Yes

Notes: Each column estimates the 2SLS specification from Table III Column (2) separately for each age group. Standard errors clustered at the county level. Data at the county-year-age level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

5.2 Robustness

Section 4.3 identified five threats to predetermination. The abortion result is stable across robustness checks (Table V). Light checks (pre-trends, leave-out stability, instrument strength, estimator range) confirm the baseline; more demanding specifications that absorb the cross-county variation identifying the IV mechanically attenuate the estimate but do not overturn it.

Figure II: Event Study: CPC Effects on Abortion, Birth, and Pregnancy Rates



Notes: County-year panel of 146 NC and SC counties, 1990–2019. Event-study coefficients on indicators for years relative to a county’s first CPC opening, estimated jointly with county and year fixed effects, the full lagged control set, and $\sqrt{\text{pop}}$ weights. Event time -1 (the year before the first opening) is the omitted reference period; event-time leads run from -10 to -2 and lags run from 0 to $+9$. Three outcomes overlaid in the same plot: log county abortion rate (navy circles), log birth rate (maroon triangles), and log pregnancy rate (green diamonds). Vertical bars are 95% confidence intervals based on standard errors clustered at the county level. Counties that never receive a CPC during the sample window contribute to the control group. The figure pools across age groups; per-age event studies are available in the replication package.

Joint F -tests of pre-treatment event study coefficients fail to reject zero pre-trends for all three outcomes (Figure II).²⁷ The pregnancy pre-trends matter for the mechanism interpretation.

²⁷ Abortion $p = 0.359$, birth $p = 0.547$, pregnancy $p = 0.446$. Although the joint F -test fails to reject for pregnancy, two of the nine individual pre-treatment coefficients are statistically significant (event times -5 and -4). Their magnitudes are small and do not trace a monotonic pre-trend, but the pattern motivates caution in interpreting the event study as mechanism evidence. The main mechanism evidence comes from the continuous-intensity IV estimates. The event study is descriptive, however, since it uses a binary first-opening treatment indicator rather than the continuous CPC-intensity measure that identifies the IV (§4.3). Identification rests on the predetermination assumption tested via residual orthogonality on the IV regressions in Appendix Table A5, not on event-study pre-trends.

Table V: Robustness Summary: Log Abortion Rate

Specification	$\hat{\beta}$	SE	First-stage F	N
Baseline 2SLS	-0.142***	(0.039)	196.1	4,376
OLS	-0.072**	(0.029)	—	4,376
TRAP + PI law controls	-0.134***	(0.037)	195.1	4,376
Stable clinic access (>30mi)	-0.149***	(0.044)	173.9	3,027
Lagged outcome control	-0.096***	(0.033)	135.9	4,080
$X_c \times$ year FE	-0.127***	(0.035)	190.0	4,376
+ Prot $\times$ Rep $\times$ year interaction	-0.131***	(0.036)	188.4	4,376
County-specific linear trends	-0.088	(0.089)	13.3	4,376
Drop border counties	-0.118**	(0.052)	55.5	2,490
Poisson-CF	-0.070*	(0.041)	—	4,380

Notes: Each row re-estimates the main 2SLS specification with one modification. The dependent variable is log county abortion rate throughout. All specifications include county and year fixed effects and the full control set unless noted otherwise. Population-weighted unless noted. Standard errors clustered by county. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The joint test cannot reject flat pregnancy pre-trends, but two of the nine individual pregnancy leads (event times -5 and -4) are significant, so the event study alone does not establish parallel pregnancy trajectories. The interpretation of the post-treatment pregnancy null as a causal zero accordingly rests primarily on the continuous-intensity IV estimates rather than the binary event study.

Three further checks address influence and instrument validity. A leave-two-county-out jackknife rules out the result being driven by a small set of influential counties: every one of the 10,585 leave-two-out estimates remains negative, 95 percent of county pairs shift the estimate by less than 8 percent, and the maximum shift is 26 percent (Online Appendix Figure [OA.5](#) and Online Appendix Table [OA.16](#)). A permutation test reassigns the instrument across counties and recovers a median first-stage F of 0.66 versus 196.1 in the actual data ($p = 0.000$; Table [A9](#)), ruling out the possibility that first-stage strength reflects overfitting to county characteristics. Oster (2019) bounds the selection on unobservables required to drive the estimate to zero at $\delta^* = 13.34$ (Online Appendix Table [OA.14](#), Online Appendix [§OA.5.5](#)): unobservable selection would need to be more than thirteen times stronger than observable selection to overturn the result.

The credible range for the abortion rate estimate across estimators is $[-0.142, -0.070]$ ([§4.4](#)), with 2SLS at the upper end (in absolute value) and the Poisson control function at the

lower end.²⁸ Augmented synthetic control estimators produce consistent negative ATTs (Online Appendix Figures OA.6 and OA.7 and Online Appendix Table OA.20; the SCPI log-abortion ATT of -0.069 is within rounding of the Poisson CF), triangulating the result across fundamentally different identification approaches.

The 2SLS coefficient remains between -0.114 and -0.149 across specifications with strong first stages (excluding the lagged-outcome controls, discussed next), varying with the control set but not with the identifying assumption. Controlling for lagged abortion rates attenuates the estimate, and much of this attenuation is mechanical: the Nickell (1981) bias bound for a fixed-effects panel with $T = 30$ is approximately 0.051 in magnitude, accounting for most of the decline from baseline to the deepest lagged-outcome specification (Appendix Table A5). A direct test confirms that no subset of lagged observables predicts the 2SLS residual ($F < 0.01$, $p > 0.99$; Appendix Table A5, Panel C), so the attenuation is consistent with bias in dynamic specifications rather than evidence against predetermination.

Adding county-specific linear trends yields a smaller and statistically insignificant estimate (-0.088 , $F = 13.3$). The first stage collapses from $F = 196.1$ to $F = 13.3$, and the Anderson-Rubin confidence set widens to $[-0.373, 0.065]$ (Table A7). The instrument is identified by cross-county nonlinear path-dependent compounding from the hazard model (§4.2.2), and county-specific trends absorb most of the residual variation after fixed effects already absorb 77 percent.²⁹ The county-trends specification is therefore a low-power diagnostic; the lagged-outcome and residual-predictability tests in the preceding paragraph are more direct checks of predetermination.

²⁸The Poisson CF analytic standard error is 0.041 ($|t| = 1.71$, $p < 0.10$). A wild cluster bootstrap (999 replications, resampling counties, Webb six-point weights) yields a bootstrap p -value of 0.058 , confirming marginal significance. The reduced-form regression of log abortion rates on the instrument yields -0.117 ($t = -3.44$; Appendix Table A11).

²⁹County-specific quadratic trends yield virtually identical results (-0.090 , $F = 12.9$; Online Appendix Table OA.10). A linear IV benchmark using the hazard covariates without the logit link or forward simulation yields a weak first stage ($F = 11.6$), confirming that identification requires the nonlinear hazard model's path-dependent compounding, a feature the Monte Carlo validation in §4.5 supports. A funding interaction test finds no evidence that CPC effects scale with organizational resources (-0.004 , s.e. = 0.053), consistent with the effect operating through the counseling encounter rather than the scale of material aid.

6 Discussion

The estimates indicate that CPCs reduce abortions at magnitudes comparable to supply-side clinic closures, but through a fundamentally different channel. Clinic closures raise the cost of obtaining an abortion; CPCs change the decision conditional on access. Three things follow from this distinction. The aggregate magnitudes need to be checked against external evidence on attendance and persuasion (§6.1). The welfare analysis depends on the substitution mechanism, marginal birth quality, the dollar scale of the forfeited abortion surplus, and the information content of counseling, not the size of an access cost (§6.2). And portability to the post-*Dobbs* environment, where CPC funding has scaled materially and abortion access has been further restricted, remains the central open policy question (§6.3). These implications share a common thread. The data are consistent with CPCs persuading women near the abortion-vs-birth decision margin, not women firmly committed to abortion. The redirected woman was already close to choosing birth, so her foregone surplus is small. This explains both the modest aggregate impact and the small per-redirected-woman welfare cost relative to per-woman averages in supply-side studies.

6.1 Magnitudes

The per-unit effect (7.0 to 14.2 percent reduction per CPC per 10,000 women) is comparable to the supply-side clinic closure estimates in Lindo et al. (2020).³⁰ CPC intensity rose from 0.21 to 0.40 per 10,000 women between 1990 and 2019, reducing aggregate county abortion rates by 2.8 percent, or 5.7 percent of the aggregate decline (1.4 percent and 2.8 percent, respectively, using the Poisson CF). This expansion implies 970 averted abortions per year across NC and SC at the 2SLS estimate (480 at the Poisson CF), at a public cost of \$1,031 per averted abortion.

These magnitudes are consistent with external evidence on CPC attendance and reported attitude change. The 970 averted abortions across 146 locations implies 6.6 changed decisions per center per year at the 2SLS estimate (3.3 at the Poisson CF), less than one per month. Service data from Ohio CPCs indicate that only 5 percent of clients report changing their views against abortion (Kissling et al., 2023). If each center serves 400 clients annually and 20 percent are abortion-

³⁰The baseline CPC estimate (-0.142) is of similar order to the supply-side estimates; the conservative $X_c \times \text{year}$ FE specification yields -0.127 (Table V).

minded, a 5 percent conversion rate yields 4 changed decisions per center per year, bracketed by the Poisson CF and 2SLS estimates. States that have tripled CPC funding since *Dobbs* are purchasing substantially larger treatment doses than what NC and SC experienced over three decades.³¹

The implied DellaVigna and Gentzkow (2010) persuasion rate provides another benchmark for the magnitude. The rate f is the share of message-exposed receivers who change behavior, adjusted for the size of the persuadable population. The implied f here is 0.21 (2SLS) or 0.10 (Poisson CF), comparable to political canvassing (12–20 percent) and larger than partisan media (3–8 percent; DellaVigna and Kaplan, 2007). Among demand-side fertility interventions, Kearney and Levine (2015) find a 4.3 percent reduction in teen births through broadcast media reaching millions of viewers, and La Ferrara et al. (2012) find a 5 percent fertility reduction from telenovela exposure in Brazil. A comparable persuasion rate from 146 physical locations is consistent with face-to-face counseling being more intensive per exposure but reaching fewer women.

6.2 Welfare

Four factors determine the welfare effect of CPC-induced substitution from abortion to birth. Substitution must occur, marginal birth quality must be measured, the dollar scale of the forfeited abortion surplus must be calibrated, and the information content of the counseling must be characterized. CPCs that correct misperceptions about adoption support or abortion risks reduce the surplus a redirected woman forfeits; CPCs that distort her beliefs increase it. The county-level data cannot pin down the fourth factor.³²

The first three factors hold among teenagers and at ages 25 and older, so the magnitude analysis proceeds at both. At ages 20–24, only a fraction of averted abortions appear as offsetting

³¹Multi-state attendance data suggest 12–20 percent of women with a pregnancy history have visited a CPC (Yang et al., 2025). Applied to NC and SC, this implies 14,000–20,000 pregnant women visiting CPCs annually, of whom 2,700–3,900 would otherwise have obtained an abortion based on the baseline abortion-to-pregnancy ratio (11.1/57.4). The implied conversion rate among abortion-minded visitors is 12 percent (Poisson CF) to 36 percent (2SLS). The lower end is consistent with modest influence on near-marginal women; the upper end reflects complier-county weighting toward mid-sized counties with higher per-capita CPC intensity.

³²Any welfare evaluation based on a treatment effect faces what might be called a *persuasion curse*. The redirected woman is selected for having perceived surplus closest to zero. If her perceived surplus differs from her true surplus because the counseling distorted her beliefs, the structural calibration provides a lower bound on the welfare cost. This parallels Bernheim and Rangel (2009) on revealed preference under distorted beliefs and the MVPF framework of Hendren and Sprung-Keyser (2020), and is consistent with the broader finding in Allcott et al. (2020) that consumption of a persuasion technology can itself shift revealed-preference valuations. The mission-awareness gap documented in §2 raises the possibility that some decisions reflect manipulated beliefs rather than informed updating.

births, with the fraction bounded above at 60% (§5.1). The per-averted-abortion welfare magnitude at 20–24 is therefore at most 60% of the substitution-only benchmark, and I do not produce a separate dollar calibration for this age.

Dollar calibrations. The public funding cost per averted abortion is straightforward. Public funding to NC and SC CPCs averaged about \$1 million annually during the study period, yielding \$1,031 per averted abortion at the 2SLS estimate.

The total social cost is larger. Public funding accounts for only 4–7 percent of total CPC resources. Adding private donations and volunteer labor valued at the QCEW average wage for social service occupations (Online Appendix §OA.4.1) puts the full social cost per averted abortion at \$16,000–\$24,000.

The private welfare cost of abortion-to-birth substitution comes from a structural calibration. I model the surplus distribution as logistic, anchored to the Turnaway Study (Foster et al., 2022), with complementary anchors from the international denied-abortion evidence (Londoño-Vélez and Saravia, 2025) and the teen dropout literature (Kearney and Levine, 2012). The implied welfare cost per redirected birth is \$418–\$2,554 (Table A13).³³ These numbers are far smaller than the per-woman welfare losses documented in the anchor papers because the structural calibration prices only the marginal woman’s surplus. By construction, the redirected woman was near abortion-vs-birth indifference at the moment of CPC contact, so her revealed surplus from abortion is small. The Turnaway and teen-mother averages pool over women with stronger preferences for abortion and correspondingly larger welfare losses. External costs (LBW outcomes, taxpayer costs) are not priced here; they enter the welfare analysis through the marginal-quality finding and the public funding cost discussed above. Under one-for-one substitution, the cost per redirected birth equals the cost per averted abortion. Comparing the moderate estimate of \$1,302 to public spending of \$1,031 implies a marginal excess burden (MEB) of 1.26; under the low scenario the MEB falls to 0.41, and under the high scenario it rises to 2.48 (Table A13). The denominator matters as much as the scenario. At the Poisson control function estimate of 480 averted

³³The J-test of the logistic distributional assumption yields $J = 0.68$, $p = 0.71$. A sufficient-statistics calibration (Chetty, 2009) using the Myers (2017) distance semi-elasticity yields a lower bound of \$59–\$787 (Table A14), but this approach prices a non-monetary persuasion mechanism using a monetary distance elasticity and likely understates the true welfare significance by an order of magnitude.

abortions per year, the public cost per averted abortion is approximately \$2,083 (\$1 million over 480 abortions), and the MEB becomes 0.20 (low), 0.63 (moderate), and 1.23 (high). Across the three scenarios and two estimators, the MEB therefore ranges from 0.20 to 2.48, with the moderate scenario above break-even at the 2SLS denominator and below it at the Poisson CF denominator. The MEB uses public funding because a key policy question is whether public dollars are well spent.

Ages 25–34. At ages 25–34 the substitution evidence is the strongest of any age group (§5.1). But the marginal-quality differential is small and statistically indistinguishable from zero, so the Wald decomposition cannot identify whether marginal births at 25–34 are better or worse than baseline. The external cost literature also provides weaker priors than for teen births, and the welfare cost per averted abortion at 25–34 is therefore not pinned down.

Teenagers. Among teenagers the analysis produces three pieces of evidence pointing toward large welfare cost.

First, substitution operates among teenagers. The count-level IV and Poisson marital decomposition establish this in §5.1, with the offsetting birth response significant and concentrated among unmarried teens.

Second, those marginal births are worse than average. The Wald decomposition of the teen LBW effect (Online Appendix Table OA.1) implies a marginal LBW rate of 12.7% against a baseline of 9.5%, with both numerator and denominator individually significant at the 1% level ($\hat{\beta}_{LBW} = 5.61, t = 3.28$).^{34,35} Teenagers are the only age group where both the substitution count

³⁴Under heterogeneous treatment effects (Imbens and Angrist, 1994), the Wald ratio identifies a feature of the joint distribution of the LBW response and the birth response across compliers, not the LBW rate among compliers in either marginal distribution. If the LBW response and the birth response are positively correlated across compliers, in the sense that compliers who induce more births also have higher-LBW pregnancies, the reported 12.7 percent is an upper bound on the marginal LBW rate. If they are negatively correlated, it is a lower bound. The qualitative conclusion that the marginal teen birth has worse health than the cell average survives either bound under the maintained assumption that complier compositions overlap across the LBW-count and birth-count regressions, which they do by construction since both use the same instrument and sample.

³⁵Applying the within-twin hospital-cost coefficients in Almond et al. (2005) and the long-run earnings effects in Black et al. (2007) to the marginal teen birth's 3.2 percentage-point LBW differential implies additional inpatient charges of \$110–\$610 and present-value earnings losses of \$400–\$900 per marginal teen birth, for an LBW-attributable welfare cost on the order of \$500–\$1,500 per marginal teen birth, or 20–60% of the structural welfare estimate of \$418–\$2,554. The dollar ranges are derived calibrations from the source coefficients, not figures reported in those papers.

and the marginal-quality differential are individually significant.³⁶ The pattern is consistent with the same selection mechanism documented in Ananat et al. (2009). There, expanded abortion access improved cohort outcomes because the women on the abortion margin had higher-risk pregnancies. CPC-induced substitution operates on the same margin in reverse, producing teen births with higher LBW than the cell average.

Third, the per-birth welfare stakes are higher among teens than other age groups. Teen mothers complete high school at lower rates, with downstream economic costs well documented (Kearney and Levine, 2012) and largest for less-advantaged teens (Lang and Weinstein, 2015).³⁷ Teenagers are therefore the subgroup where the welfare cost concentrates across information-quality scenarios.

6.3 External Validity and the Post-*Dobbs* Environment

North Carolina and South Carolina had legal abortion access combined with parental consent and waiting periods throughout the study period, a regulatory environment shared by most states in the pre-*Dobbs* South.³⁸ The CPC organizational landscape is nationally representative. Over 75 percent of U.S. CPCs affiliate with the same networks (Care Net, Heartbeat International, NIFLA) that operate locally, and the NC CPC-to-clinic ratio of 6:1 exceeds the national 3.4:1, so the treatment intensity if anything overstates the typical state. The results generalize most directly to states that shared this pre-*Dobbs* policy and organizational environment, consistent with the finding in Dehejia et al. (2021) that macro covariates drive treatment-effect heterogeneity across fertility experiments.

The IV identification structure introduces a second scope condition. With a continuous instrument and continuous endogenous variable, 2SLS estimates a weighted average of marginal causal responses, with counties where actual CPC presence deviates most from the simulated

³⁶The marital-specific decomposition (Online Appendix Table OA.2) tells a consistent story. Unmarried teenagers show marginal LBW of 11.3 percent versus a baseline of 9.5 percent ($\beta_{\text{births}} = 58.1$, $t = 2.35$). At ages 30–34, the LBW share worsens by 0.9 percentage points ($p < 0.01$), but the Wald denominator is uninformative ($t = -1.29$), so the source of the share worsening cannot be identified.

³⁷Noghanibehambari et al. (2025) document multigenerational birth-outcome consequences of abortion access, with mothers' early-life exposure to legal abortion improving the next generation's birth outcomes through maternal educational attainment and prenatal care. The data here speak to one-generation marginal quality only.

³⁸Both states were classified as "hostile" by the Guttmacher Institute by 2019, in a cluster of 29 states with moderate-to-restrictive policy environments. The two states are slightly older, slightly less educated, and substantially more religious than the national average (ACS 2019–2023; Pew Research Center (2025)).

instrument contributing most to identification (Table A10 in Appendix E.1). The top leverage quartile has lower baseline abortion rates (12.2 vs. 14.3 per 1,000), similar birth rates, and a higher birth-to-abortion ratio at ages 25–29 (7.0 vs. 5.7). These are predominantly mid-sized counties (10,000–30,000 women) in the 1990s and early 2000s; metro counties with 90,000+ women contribute little because per-capita CPC intensity is low. In mid-sized counties, a single CPC serves a smaller population, generating higher per-capita exposure and a larger percentage effect on the abortion rate (Online Appendix Figures OA.3–OA.4; see Angrist and Pischke, 2009, for a general discussion of LATE heterogeneity with continuous instruments). The effect per CPC may therefore be smaller in large metropolitan areas where each center serves a larger and more dispersed catchment.

Direct evidence on post-*Dobbs* portability comes from a sample restriction reported in §4.3. Restricting to counties always more than 30 miles from the nearest abortion provider during the study period yields an IV estimate of similar magnitude to the full-sample baseline (−0.149 vs. −0.142; first-stage $F = 173.9$; Online Appendix Table OA.11). These are counties where the access cost to abortion was already high, providing one pre-*Dobbs* analogue to high-access-cost environments. The estimated effect is therefore not driven by counties with nearby abortion access, supporting portability to settings where abortion is already costly to obtain.

The post-*Dobbs* environment is qualitatively different. During the study period, CPCs operated as demand-side influences competing with legal access. In the 18 ban states, the outside option is now interstate travel for the roughly 170,000 patients who crossed state lines for abortion care in 2023, and the demand-side persuasion margin may be smaller when the default is already non-abortion. Dench et al. (2024) estimate that ban states experienced a 2.3 percent birth increase relative to non-ban states, about 30,000 additional births annually from supply-side restriction alone. Joyce et al. (2013) earlier projected that a 31-state ban would produce up to a 4.2 percent national birth increase from distance-driven access loss alone. The post-*Dobbs* funding scale-up (§2.2) means CPCs now operate at materially larger doses than during the study period.³⁹ Whether these larger doses generate proportionally larger effects, or whether the demand-side

³⁹Restriction-driven distance changes also produce welfare consequences beyond fertility. Aslim et al. (2025) document increased economic hardship and financially motivated crime following Texas clinic closures, and Dave et al. (2025) find post-*Dobbs* increases in intimate partner violence.

channel attenuates without a meaningful abortion outside option, is the central post-*Dobbs* policy question. Taken together, the three subsections show that CPCs achieved real abortion reductions at modest aggregate doses with bracketable welfare costs in the pre-*Dobbs* environment; whether these properties survive at materially larger doses and with a contracted outside option is the question the next section returns to.

7 Conclusion

One additional CPC per 10,000 women reduces abortion rates by 7.0 to 14.2 percent, comparable in magnitude to supply-side clinic closures but operating through a fundamentally different channel. Clinic closures raise the cost of access; CPCs change the decision conditional on access. The aggregate pregnancy rate response is small and not statistically distinguishable from zero, ruling out full deterrence as the dominant mechanism. The mechanism is substitution from abortion to birth, with partial deterrence at moderate magnitudes consistent with the data but not separately identified. At ages 25 and older, averted abortions appear as additional births at the aggregate level. Count-level evidence supports substitution among teenagers. At ages 20–24, the substitution share is bounded above at 60 percent. The data are consistent with the CPC effect concentrating on women near the abortion-vs-birth indifference margin, which would reconcile the per-unit magnitude with the small per-redirected-woman welfare cost.

The welfare implications vary by age. Among teenagers, CPC-induced births have low birth weight rates 3.2 percentage points above baseline, making this the subgroup where welfare concerns are most empirically grounded. At ages 25–34, substitution is clear but the marginal-quality and information-quality factors are both unidentified at this age, so the welfare cost cannot be pinned down. At ages 20–24, abortions decline but the birth response is statistically indistinguishable from zero and bounds the substitution share at no more than 60%; the mechanism at this age is partially identified rather than resolved.

Because CPCs operate on the demand side, their effects depend on the availability of abortion as a meaningful outside option. In the pre-*Dobbs* environment studied here, women who encountered CPCs had a choice; the CPC shifted the margin for those near indifference. Whether the \$489 million in post-*Dobbs* CPC funding generates the same effects in states with near-total

bans, where treatment doses are materially larger and the outside option has become much more difficult to access, is the central open policy question. Individual-level data linking CPC contact to pregnancy decisions and birth outcomes would resolve the information-quality question that county-level data leave open.

The forward-simulation IV approach developed here generalizes beyond CPCs. Wherever facility or program placement is endogenous and path-dependent without a natural experiment to exploit, a discrete-time hazard model of entry estimated from predetermined area characteristics can generate a valid instrument through the same simulation logic. Candidate applications include charter school placement, opioid treatment facility location, broadband infrastructure roll-out, and other locally-decided services with strategic placement and path-dependent expansion.

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A Instrument Construction and Hazard Model

A.1 Renewal Framework for Multiple Openings

Table A1: Variable Treatment in the Forward Simulation

Category	Variables	Rationale
<i>Updated within simulation</i>	CPC count, distance to nearest CPC, polynomial time trend	Endogenous to simulated expansion process
<i>Held at observed values</i>	Demographics, religious composition, political ideology, unemployment, distance to nearest abortion provider	Evolve in data but treated as exogenous to CPC openings within each draw
<i>Frozen at 1990</i>	Lagged abortion rates	Avoids embedding assumed treatment effects

Notes: “Updated” variables change recursively within each of the $R = 1,000$ simulation draws as new CPCs open. “Observed” variables enter at their actual panel values in each county-year. “Frozen” variables are fixed at 1990 cross-sectional values throughout the simulation.

Counties can receive multiple CPCs over the sample period. I extend the single-spell hazard to a renewal framework following Mroz (2012). The renewal decomposition expresses the count distribution as a sequence of independent binary events, each estimable by logit regardless of the true count-generating process. State dependence enters through $\text{CPC}_{c,t-1}$, so the hazard at each step conditions on accumulated stock. The outcome $\text{NewCPC}_{c,t}$ resets to zero each year and can equal one multiple times for the same county across t , with the likelihood taking the standard pooled logit form:

$$\mathcal{L} = \prod_c \prod_t p_{c,t}^{\text{NewCPC}_{c,t}} (1 - p_{c,t})^{1 - \text{NewCPC}_{c,t}} \quad (8)$$

This structure allows the hazard model to capture both first entries and subsequent CPC additions, with the existing stock entering as a time-varying covariate that governs the rate of accumulation.

A.2 Forward Simulation Algorithm

For each county c and target year t , I generate $R = 1,000$ simulated CPC expansion paths:

1. **Initialize:** Set $\widehat{\text{CPC}}_{c,1990}^{(r)} = \text{CPC}_{c,1990}$ (observed initial state) for all r .
2. **Forward simulate:** For each draw $r = 1, \dots, R$ and year $\tau = 1991, \dots, t$:
 - (a) Compute predicted opening probability $\hat{p}_{c\tau}^{(r)}$ using simulated lagged CPC counts, distances, and time trends, with all other covariates held at their observed values;
 - (b) Draw $U_{c\tau}^{(r)} \sim \text{Uniform}(0, 1)$;
 - (c) Open a CPC if $U_{c\tau}^{(r)} \leq \hat{p}_{c\tau}^{(r)}$;
 - (d) Update cumulative count, distance to nearest CPC, and polynomial time trend.

A.3 Results from the CPC Logit Model

Table A2 presents estimates from the hazard model. Column (1) provides the baseline specification used to construct instruments. An existing CPC reduces the probability of additional

openings in the same county, indicating market saturation or coordination to avoid duplication. Greater driving distance to the nearest CPC increases the probability of a new opening, confirming CPCs target underserved areas.

Table A2: Predicting the Opening of Crisis Pregnancy Centers

	(1)	(2)
No. of CPCs (lagged)	-0.050 (0.017)	-0.057 (0.016)
Nearest CPC (lagged)	0.001 (0.001)	0.001 (0.001)
Nearest Clinic (lagged)	-0.000 (0.000)	-0.000 (0.000)
Abortion rate per 1000 women 10–44 (lagged)	0.001 (0.003)	-0.001 (0.004)
Log population (lagged)	0.000 (0.000)	0.000 (0.000)
Pop. share age 10–19 (lagged)	0.072 (0.553)	0.018 (0.603)
Non-white share (lagged)	0.186 (0.554)	0.386 (0.605)
Unemployment rate (lagged)	0.001 (0.003)	0.004 (0.006)
U.S. GOP vote share (lagged)	-0.068 (0.037)	-0.076 (0.038)
Protestant share (lagged)	0.113 (0.224)	0.174 (0.187)
N	2,129	1,987
County FE	Yes	Yes
Time Trend (3rd degree)	Yes	No
Year FE	No	Yes

Notes: Estimates from the renewal model of CPC location choice. All independent variables lagged one period. Average marginal effects via delta method; each coefficient is a percentage-point change in the probability of a new CPC opening. Standard errors in parentheses.

B Formal Identification

B.1 Two-Assumption Decomposition

The main text’s predetermination assumption can be decomposed into two conditions.

Assumption 1 (Conditional Orthogonality): Actual organizational timing shocks are mean-independent of unobserved fertility shocks, conditional on observable history and fixed effects:

$$\varepsilon_{c,t}^{\text{DGP}} \perp u_{c,t} \mid \mathcal{F}_{c,t-1}, \mu_c, \lambda_t \quad (9)$$

where $\mathcal{F}_{c,t-1} = \{W_{c,\tau}\}_{\tau=1}^{t-1}$ denotes the historical information set.

Assumption 2 (Predetermination): Observable county characteristics are predetermined

with respect to current-period fertility shocks:

$$\mathbb{E}[u_{c,t} \mid W_{c,t-1}, \mu_c, \lambda_t] = 0. \quad (10)$$

Since the instrument converges to $\mathbb{E}[\text{CPC}_{c,t-1} \mid W_{c,t-1}]$ and is measurable with respect to $W_{c,t-1}$, predetermination directly establishes the exclusion restriction via the law of iterated expectations.

C Asymptotic Properties of the Simulated Instrument

This appendix formalizes the asymptotic argument for the forward-simulation IV. The main text establishes the economic content of the predetermination assumption and validates the estimator via Monte Carlo. Here I show that consistency and asymptotic normality follow from standard results on generated instruments and simulation-based estimation, making explicit what the citations to Amemiya (1974), Murphy and Topel (1985), and Pakes and Pollard (1989) imply for the specific estimator used in this paper.

C.1 Setup and Notation

Let $\hat{\beta}$ denote the logit MLE from the hazard model (equation 2), estimated on C counties over T periods. For a given parameter vector β and county-year pair (c, t) , the forward simulation generates S independent CPC expansion paths $\{Z_{c,t}^{(s)}(\beta)\}_{s=1}^S$ by drawing i.i.d. uniform shocks and recursively updating CPC counts through the logit hazard. Define:

$$\bar{Z}_{c,t}(\beta) = \frac{1}{S} \sum_{s=1}^S Z_{c,t}^{(s)}(\beta) \quad (11)$$

$$Z_{c,t}^*(\beta) = \mathbb{E}_U[Z_{c,t}^{(1)}(\beta)] \quad (12)$$

where the expectation in (12) is over the simulation draws U for fixed β and fixed observables $W_{c,t-1}$. The instrument used in estimation is $\bar{Z}_{c,t}(\hat{\beta})$; the population instrument is $Z_{c,t}^*(\beta^*)$, where β^* is the probability limit of the logit MLE. Note that β^* is the best logit approximation to the true hazard, not necessarily the true data-generating process—the Monte Carlo validation in §4.5 confirms that instrument validity does not require correct specification.

C.2 Convergence of the Simulated Instrument

Claim 1 (Simulation convergence). For fixed β and fixed observables $W_{c,t-1}$,

$$\bar{Z}_{c,t}(\beta) \xrightarrow{a.s.} Z_{c,t}^*(\beta) \quad \text{as } S \rightarrow \infty.$$

Proof sketch. The simulation draws $\{U_{c,\tau}^{(s)}\}_{\tau,s}$ are i.i.d. across s conditional on β and $W_{c,t-1}$. Each path $Z_{c,t}^{(s)}(\beta)$ is a bounded function of these draws (CPC counts are bounded above by T and below by the initial stock). The strong law of large numbers for i.i.d. bounded random variables gives pointwise almost-sure convergence. $\square$

Claim 2 (Hazard consistency). Under standard regularity conditions for MLE—compactness of the parameter space, identification of β^* , and bounded score— $\hat{\beta} \xrightarrow{p} \beta^*$ as $C \rightarrow \infty$ with T fixed.

This is the standard consistency result for the logit MLE (Amemiya, 1985, Chapter 4). The key point is that β^* is the pseudo-true parameter—the probability limit of the MLE under possible misspecification—not the parameter of the true data-generating process. The exclusion restriction depends on predetermination of the hazard covariates $W_{c,t-1}$, not on the hazard being correctly specified.

Claim 3 (Joint convergence). If $Z_{c,t}^*(\beta)$ is continuous in β at β^* , then

$$\bar{Z}_{c,t}(\hat{\beta}) \xrightarrow{p} Z_{c,t}^*(\beta^*) \quad \text{as } C, S \rightarrow \infty.$$

Proof sketch. Write

$$\bar{Z}_{c,t}(\hat{\beta}) - Z_{c,t}^*(\beta^*) = \underbrace{[\bar{Z}_{c,t}(\hat{\beta}) - Z_{c,t}^*(\hat{\beta})]}_{\text{simulation error}} + \underbrace{[Z_{c,t}^*(\hat{\beta}) - Z_{c,t}^*(\beta^*)]}_{\text{estimation error}}.$$

The first term vanishes as $S \rightarrow \infty$ by Claim 1, uniformly in β in a neighborhood of β^* (the bounded convergence argument extends uniformly over compact parameter sets; see Newey and McFadden, 1994). The second term vanishes as $C \rightarrow \infty$ by the continuous mapping theorem applied to Claim 2. Joint convergence follows by the triangle inequality. $\square$

The continuity condition is satisfied because the logit link is smooth and the recursive simulation maps smoothly differentiable hazard probabilities to CPC counts through a finite sequence of Bernoulli draws.

C.3 Consistency of the 2SLS Estimator

The 2SLS estimator of γ in equation (7) uses $\bar{Z}_{c,t}(\hat{\beta})$ as instrument for $\text{CPC}_{c,t}$. Consistency requires two conditions:

Relevance. $\text{Cov}(Z_{c,t}^*(\beta^*), \text{CPC}_{c,t} \mid X_{c,t}, \mu_c, \lambda_t) \neq 0$. Verified empirically: the first-stage F -statistic is 170.

Exclusion. $\mathbb{E}[u_{c,t} \cdot Z_{c,t}^*(\beta^*) \mid \mu_c, \lambda_t] = 0$. This follows from two observations. First, $Z_{c,t}^*(\beta^*)$ is a measurable function of $W_{c,t-1}$: for any fixed β , the population instrument is a deterministic function of the hazard covariates, computed by integrating out the simulation draws. Second, the predetermination assumption (equation 4) states $\mathbb{E}[u_{c,t} \mid W_{c,t-1}, \mu_c, \lambda_t] = 0$. The exclusion restriction then follows by the law of iterated expectations, as established in equation (5) of the main text.

The critical distinction from generated instruments in demand estimation (e.g., Pakes and Pollard, 1989) is that the instrument here is a function of *predetermined* variables. In BLP-style applications, simulated instruments may depend on endogenous market structure, requiring a separate exclusion restriction argument for each component. Here, the instrument inherits exogeneity directly from predetermination regardless of whether the hazard model is correctly specified. This is the Amemiya (1974) point: any function of exogenous variables is a valid instrument, so replacing the known function $Z^*(\beta^*)$ with the estimated $\bar{Z}(\hat{\beta})$ does not affect the exclusion restriction in the population limit.

Claim 4 (Consistency). Under relevance, exclusion, Claim 3, and standard rank conditions for 2SLS, $\hat{\gamma} \xrightarrow{p} \gamma$ as $C, S \rightarrow \infty$.

Proof sketch. The 2SLS estimator can be written as

$$\hat{\gamma} = \gamma + \left(\frac{1}{N} \sum_{c,t} \bar{Z}_{c,t}' \widehat{\text{CPC}}_{c,t} \right)^{-1} \left(\frac{1}{N} \sum_{c,t} \bar{Z}_{c,t}' u_{c,t} \right)$$

where $N = C \times T$ and both $\bar{Z}$ and $\widehat{CPC}$ are residualized on $X_{c,t}$, μ_c , and λ_t . By Claim 3, $\bar{Z}_{c,t}(\hat{\beta}) \rightarrow Z_{c,t}^*(\beta^*)$ uniformly, so the sample moments converge to their population counterparts. The numerator converges to $\text{Cov}(Z^*, u) = 0$ by the exclusion restriction, and the denominator to $\text{Cov}(Z^*, CPC) \neq 0$ by relevance. Standard arguments then give consistency (see Newey and McFadden, 1994, Chapter 36, for the general framework). $\square$

C.4 Asymptotic Normality

Claim 5 (Asymptotic distribution). Under the conditions of Claim 4, with inference based on C clusters:

$$\sqrt{C}(\hat{\gamma} - \gamma) \xrightarrow{d} N(0, V)$$

where V is the cluster-robust sandwich variance estimator.

Two sources of estimation uncertainty could affect inference beyond the standard 2SLS variance: (i) hazard model estimation error ($\hat{\beta}$ versus β^*), and (ii) simulation noise ($\bar{Z}$ versus Z^*).

Hazard estimation error. The Murphy and Topel (1985) correction adds to the asymptotic variance a term proportional to the covariance between the second-stage moment and the hazard score. In the present setting, this correction is $O(1/C)$ and enters the same asymptotic expansion as the standard variance. Because the cluster-bootstrap reported in the paper resamples counties, re-estimates the hazard, and re-simulates within each bootstrap draw, it directly accounts for this source of uncertainty without requiring the analytic Murphy-Topel correction. The similarity of bootstrapped and analytic standard errors across all specifications (reported in the main text) confirms that the generated-regressor adjustment is quantitatively negligible.

Simulation noise. With S fixed as $C \rightarrow \infty$, the simulation error $\bar{Z}_{c,t} - Z_{c,t}^*$ contributes an additional variance term of order $O(1/S)$ to the finite-sample variance of $\hat{\gamma}$. With $S = 1,000$, this contribution is bounded at $1/1,000 = 0.001$ of the signal variance, negligible relative to the $O(1/C)$ sampling variance with $C = 146$ counties. Formally, if $S \rightarrow \infty$ at any rate as $C \rightarrow \infty$, the simulation noise is asymptotically negligible and the standard cluster-robust variance is valid (Pakes and Pollard, 1989, Theorem 3.1).

The cluster-robust sandwich estimator $\hat{V}$ is therefore consistent for V under the standard conditions: a large number of independent clusters ($C = 146$), within-cluster dependence of unrestricted form, and the generated-regressor and simulation contributions vanishing relative to the cluster-level sampling variance. Because the bootstrap directly re-estimates both the hazard and the simulation, it provides a finite-sample check on these asymptotic arguments.

C.5 Connection to Monte Carlo Validation

The formal argument shows that consistency requires only predetermination and relevance, not correct specification of the hazard model. The Monte Carlo exercises in §4.5 confirm this empirically: across nine cells of a 3×3 factorial design that deliberately misspecifies the hazard, the IV recovers the true causal parameter with bias below 0.07, 95% confidence interval coverage near nominal, and no evidence of size distortion. The Monte Carlo further demonstrates that the simulation-averaging device ($S = 1,000$) suffices for the first-stage F -statistic to stabilize—the Monte Carlo first-stage F of 74 (lower than the actual $F = 170$ due to the more severe endogeneity in the designed DGP) is well above conventional weak-instrument thresholds in all cells. Together, the formal argument and the Monte Carlo evidence provide complementary support for the estimator: the asymptotics establish *why* the estimator works (predetermination plus the law of iterated expectations), and the Monte Carlo shows *that* it works in finite samples under

conditions calibrated to exceed the severity of the empirical application.

As a practical diagnostic, the 2SLS coefficient is -0.1438 with $S = 1,000$ simulation draws, -0.1441 with $S = 500$, and -0.1435 with $S = 100$; the first-stage F -statistic is 170, 168, and 161, respectively. The coefficient is stable to the third decimal across a tenfold reduction in S , confirming that simulation noise is negligible relative to sampling variance at the implemented $S = 1,000$.

D Monte Carlo Validation

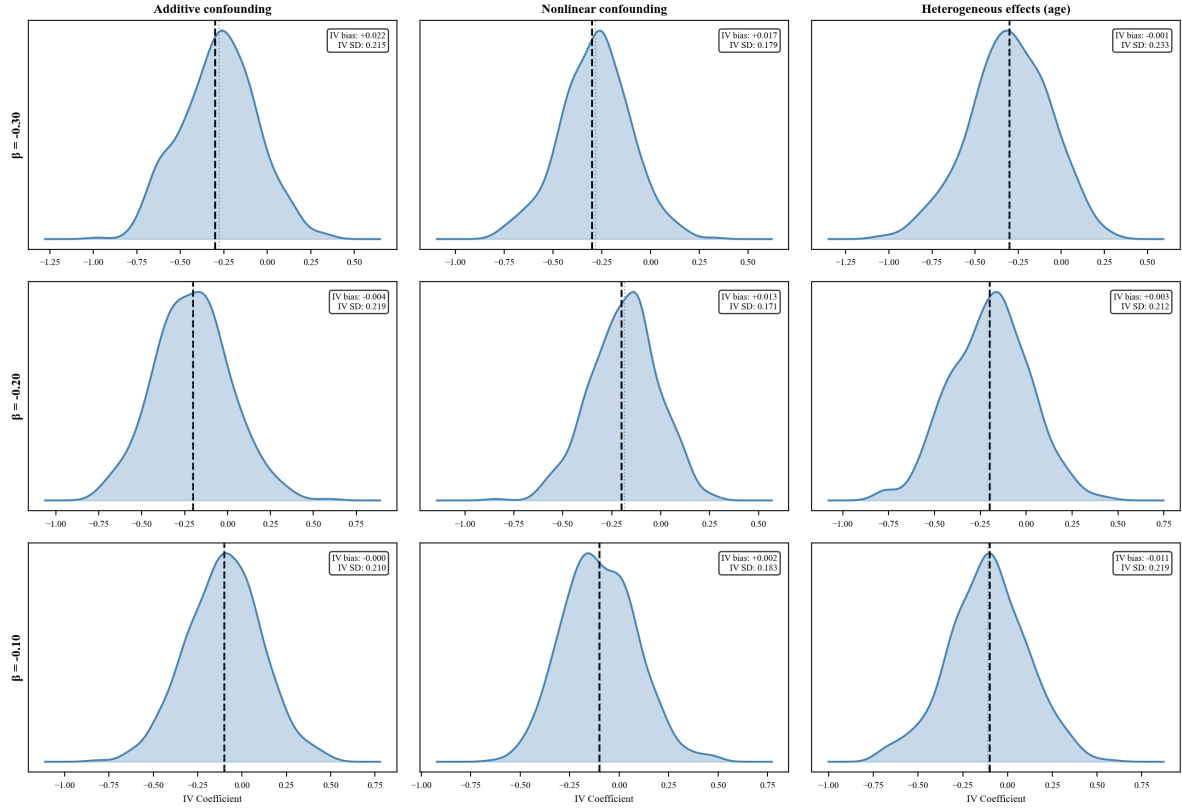
The 3×3 factorial design crosses three true treatment effects ($\beta \in \{-0.10, -0.20, -0.30\}$) with three DGP specifications (additive confounding; nonlinear confounding in which the confounder enters as its square; heterogeneous effects by county age composition), each repeated for 500 replications. The hazard model is deliberately misspecified.

Table A3: Monte Carlo Validation: Summary Statistics Across DGP Specifications

True β	DGP	Mean OLS	Mean IV	Bias OLS	Bias IV	SD(IV)	Mean F	Reps
-0.30	Additive confounding	+1.446	-0.278	+1.746	+0.022	0.216	74	500
	Nonlinear confounding	-0.251	-0.283	+0.049	+0.017	0.179	74	500
	Heterogeneous effects (age)	+1.455	-0.301	+1.755	-0.001	0.233	74	500
-0.20	Additive confounding	+1.523	-0.204	+1.723	-0.004	0.219	74	500
	Nonlinear confounding	-0.162	-0.187	+0.038	+0.013	0.171	74	500
	Heterogeneous effects (age)	+1.532	-0.197	+1.732	+0.003	0.212	74	500
-0.10	Additive confounding	+1.609	-0.100	+1.709	-0.000	0.210	74	500
	Nonlinear confounding	-0.082	-0.098	+0.018	+0.002	0.183	74	500
	Heterogeneous effects (age)	+1.605	-0.111	+1.705	-0.011	0.219	74	500

Notes: Each cell reports summary statistics from Monte Carlo replications. True β is the known treatment effect in the synthetic DGP. "Additive confounding" uses linear, additive confounding. "Nonlinear confounding" replaces Z with Z^2 . "Heterogeneous effects (age)" introduces treatment effect heterogeneity scaled by county age composition (share aged 20–29). Mean OLS uses the confounded treatment variable; Mean IV uses the forward-simulation instrument. Bias = Mean estimate – True β . F is the first-stage Kleibergen-Paap F -statistic.

Figure A1: Monte Carlo IV Validation: Kernel Density of IV Estimates



Notes: Each panel shows the Epanechnikov kernel density of the 2SLS coefficient on the simulated CPC instrument across 500 Monte Carlo replications, for one of nine design cells. Rows index the true treatment effect ($\beta \in \{-0.10, -0.20, -0.30\}$) and columns index the data-generating process (DGP): additive confounding, nonlinear confounding in which the confounder enters as its square, and heterogeneous effects scaled by county age composition. Each replication draws a synthetic outcome from the DGP and re-runs the full pipeline (hazard logit, 1,000-draw forward simulation, county-year-clustered 2SLS) using a deliberately misspecified hazard. Dashed vertical line marks the true β in each panel; dotted vertical line marks the cross-replication mean IV estimate. Cross-replication bias is below 0.07 in all nine cells; coverage of 95% confidence intervals is near nominal across the design.

D.1 Sensitivity to Predetermination Violations

The preceding design validates the IV under hazard misspecification with predetermination satisfied. Table A4 tests what happens when predetermination *fails*: the structural error includes an omitted lagged-outcome component $u_{c,t} = \phi \cdot \bar{r}_{c,t-1}^{\text{abort}} + \varepsilon_{c,t}$, where $\bar{r}_{c,t-1}^{\text{abort}}$ is the lagged county abortion rate. Since $\bar{r}_{c,t-1}^{\text{abort}}$ enters the hazard model used to construct the instrument, the exclusion restriction fails when $\phi > 0$. The violation intensity ϕ is varied from 0 (no violation) to 0.40 (extreme). At $\phi = 0$, the IV recovers the true $\beta = -0.20$ with 4.8% bias and 92% coverage—confirming the baseline results. As ϕ increases, bias grows monotonically: -34% at $\phi = 0.05$, -74% at $\phi = 0.10$, and -304% at $\phi = 0.40$. Coverage deteriorates correspondingly, from 92% to 3%.

Crucially, the predetermination diagnostic—the joint F -test of lagged observables predicting the 2SLS residual, replicating Panel C of Table A5—detects violations at every magnitude tested. At $\phi = 0.05$ (the mildest violation), the test rejects in 98% of replications. At $\phi \geq 0.10$, it rejects in 100%. At $\phi = 0$ (no violation), the rejection rate is 5%, matching the nominal size. The empirical feedback magnitude in the data—estimated by regressing the structural residual on lagged abortion rates—corresponds to $\phi \approx 0.11$, which falls in the range where the MC shows substantial IV bias (-74%) and 100% diagnostic detection. That the paper’s predetermination test does not reject in the actual data ($F < 0.01$, $p > 0.99$; Table A5, Panel C) indicates that the county and year fixed effects plus time-varying controls in the structural equation absorb the persistence that this calibration captures, leaving the identifying variation clean.

Table A4: IV Performance Under Predetermination Violations

ϕ	Mean $\hat{\beta}_{\text{IV}}$	Bias (%)	RMSE	Coverage	Mean F	Mean $\hat{\beta}_{\text{OLS}}$	Predet. test rej.
0.00	-0.190	4.8	0.093	0.92	75.4	-0.193	0.05
0.05	-0.267	-33.6	0.107	0.90	75.4	-0.217	0.98
0.10	-0.348	-73.9	0.164	0.70	75.4	-0.227	1.00
0.20	-0.523	-161.4	0.334	0.26	75.4	-0.256	1.00
0.40	-0.808	-304.1	0.614	0.03	75.4	-0.316	1.00
True β				-0.200			
Replications				100			

Notes: Each row reports IV and OLS results from 100 Monte Carlo replications at a given predetermination violation intensity ϕ . The true treatment effect is $\beta = -0.20$ throughout. The violation is modeled as an omitted lagged-outcome component in the structural error: $u_{c,t} = \phi \cdot \bar{r}_{c,t-1}^{\text{abort}} + \varepsilon_{c,t}$, where $\bar{r}_{c,t-1}^{\text{abort}}$ is the lagged county abortion rate. Since $\bar{r}_{c,t-1}^{\text{abort}}$ enters the hazard model used to construct the instrument, the exclusion restriction $\mathbb{E}[u_{c,t} \cdot \text{IV}_{c,t}] = 0$ fails when $\phi > 0$. The final column is the fraction of replications where a joint F -test of lagged observables predicting the 2SLS residual rejects at the 5% level—replicating the diagnostic in Table A5, Panel C. Coverage is the fraction of 95% confidence intervals containing the true β . RMSE is $\sqrt{\text{mean}[(\hat{\beta} - \beta)^2]}$.

E Predetermination Tests and First Stage Diagnostics

Table A5 extends the progressive lagged-outcome controls through the fourth lag and reports the Nickell (1981) bias bound. The coefficient stabilizes in the deeper-lag specifications at -0.05 to -0.07 . The Nickell bias bound of approximately 0.051 in magnitude accounts for most of the attenuation from the baseline -0.142 to the deepest lagged-outcome specification, confirming that the decline reflects incidental-parameter bias from including lagged dependent variables in a fixed-effects panel ($T = 30$), not a predetermination violation. Panel B reports serial correlation tests, with a corrected note explaining why the significant AR(2) does not carry the Arellano-Bond diagnostic interpretation in this levels specification. Panel C confirms that no subset of lagged controls predicts the 2SLS residual.

Table A5: Predetermination Tests: Progressive Lagged Outcomes and Serial Correlation

	Panel A: IV Coefficient with Progressive Lag Controls				
	Baseline (1)	+ y_{t-2} (2)	+ y_{t-2}, y_{t-3} (3)	+ $y_{t-1}, \dots, y_{t-3}$ (4)	+ $y_{t-2}, \dots, y_{t-4}$ (5)
CPCs per 10,000	-0.142*** (0.039)	-0.096*** (0.033)	-0.072** (0.032)	-0.052** (0.026)	-0.056 (0.034)
Lagged outcomes	None	y_{t-2}	y_{t-2}, y_{t-3}	$y_{t-1}, y_{t-2}, y_{t-3}$	$y_{t-2}, y_{t-3}, y_{t-4}$
Observations	4,376	4,080	3,931	3,927	3,782
First-stage F	196.1	135.9	104.2	105.8	74.8
Nickell bias bound	≈ 0.051				
	Panel B: Serial Correlation in 2SLS Residuals				
AR(1): $\hat{\rho}_1$	0.405			$p < 0.001$	
AR(2): $\hat{\rho}_2$	0.328			$p < 0.001$	
Joint AR(1,2) F -test	$F = 79.8$			$p < 0.001$	
	Panel C: Residual Predictability ($W_{c,t-1}$ on $\hat{u}_{c,t}$)				
Joint F -test	$F < 0.01$			$p > 0.99$	
Within R^2	0.0000				

Notes: Panel A re-estimates the baseline 2SLS adding progressively deeper lagged log abortion rates as controls. The coefficient stabilizes between three and five lags. The Nickell (1981) bias bound of ≈ 0.051 accounts for most of the attenuation from baseline to the five-lag specification, indicating that the decline reflects incidental-parameter bias from including lagged dependent variables in a fixed-effects panel ($T = 30$), not a predetermination violation. Panel B tests serial correlation in the baseline 2SLS residuals by regressing $\hat{u}_{c,t}$ on $\hat{u}_{c,t-k}$ with county and year FE. AR(1) captures mechanical persistence in levels residuals. The significant AR(2) does not carry the Arellano-Bond diagnostic interpretation: the Arellano-Bond AR(2) test applies to first-differenced GMM residuals, where AR(2) tests for AR(1) in the idiosyncratic error. In the levels 2SLS specification used here, AR(2) reflects the natural propagation of outcome persistence through second-order lags and does not indicate a separate predetermination violation. Panel C provides the appropriate predetermination diagnostic: regressing the 2SLS residual on all lagged controls $W_{c,t-1}$. Under predetermination, no subset should predict the residual. The joint F -test cannot reject this null ($p > 0.99$). All specifications use $\sqrt{\text{pop}}$ weights and county-clustered SEs. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A6 reports Anderson-Rubin tests for specifications with weak first stages. With a first-stage F of 196.1, the AR confidence set for the baseline specification is virtually identical to the standard Wald CI, confirming negligible weak-instrument distortion.

Table A6: Anderson-Rubin Tests for Weak-Instrument Specifications

Specification	IV Coef.	First-stage F	RF t -stat	RF p -value
County linear trends	-0.088	13.3	-1.10	0.273
Pre-2003 subsample	-0.142	12.6	-2.76	0.007
De-noised instrument	-0.275	12.8	-3.76	0.000
Baseline (comparison)	-0.142	196.1	-3.45	0.001

Notes: County-year panel of 146 NC and SC counties, 1990–2019 ($N \approx 4,376$). Dependent variable is the log county abortion rate per 1,000 women aged 10–44. For the null $H_0: \beta = 0$, the Anderson-Rubin test is equivalent to testing whether the reduced-form coefficient (outcome regressed directly on the instrument) is zero; this test is robust to weak identification. Columns report the 2SLS IV coefficient, the first-stage Kleibergen-Paap F -statistic, the reduced-form t -statistic, and the associated p -value. All rows are estimated from the production simulated instrument, so the baseline row reproduces Table III exactly. Specifications with first-stage F below the Stock-Yogo 10% threshold (16.38) are the primary targets; “Baseline (comparison)” is included as the strong-IV anchor. “County linear trends” adds county-specific linear time trends; “Pre-2003 subsample” restricts to 1990–2002; “De-noised instrument” re-projects the simulated instrument on county and year fixed effects before second-stage estimation. All specifications include county and year fixed effects, the full lagged-control set, and $\sqrt{\text{pop}}$ weights; standard errors clustered at the county level.

Table A7 extends the Anderson-Rubin analysis to the county-specific linear trends specification, where the first-stage F drops to 13.3. The AR 95% confidence set ($[-0.373, 0.065]$) is wider than the baseline ($[-0.219, -0.063]$) and includes zero, consistent with the first-stage collapse absorbing identifying variation.

Table A7: Anderson-Rubin Confidence Sets: Baseline vs. County Trends

	Baseline	County trends
2SLS point estimate	-0.142 (0.039)	-0.088 (0.089)
Wald 95% CI	$[-0.219, -0.065]$	$[-0.262, 0.086]$
AR 95% CS	$[-0.219, -0.063]$	$[-0.373, 0.065]$
Width ratio (AR/Wald)	1.014	1.258
First-stage F	196.1	13.3
N	4,376	4,376

Notes: The Wald 95% CI uses standard 2SLS inference. The Anderson-Rubin 95% confidence set inverts the AR test over a grid of candidate values and is valid regardless of instrument strength. Baseline absorbs county and year fixed effects with time-varying controls. County trends additionally absorbs county-specific linear time trends. Both columns use the production simulated instrument from Table III; the baseline column reproduces the Table III 2SLS estimate exactly. Standard errors clustered at the county level.

Table A8 reports Driscoll-Kraay standard errors, which are robust to arbitrary cross-sectional and temporal dependence, alongside the baseline county-clustered standard errors. The Driscoll-Kraay SEs on the 2SLS abortion coefficient are smaller than the county-clustered SEs by 30 to 40 percent. The pattern reflects that the Driscoll-Kraay estimator’s cross-sectional aggregation removes correlation that county clustering treats as within-cluster variance, and is

consistent with negative spatial correlation in the residuals. Under either explanation, county clustering is the conservative reporting choice for inference. The baseline inference reported throughout the paper uses county-clustered standard errors and is the more cautious of the two.

Table A8: Clustered vs. Driscoll-Kraay Standard Errors

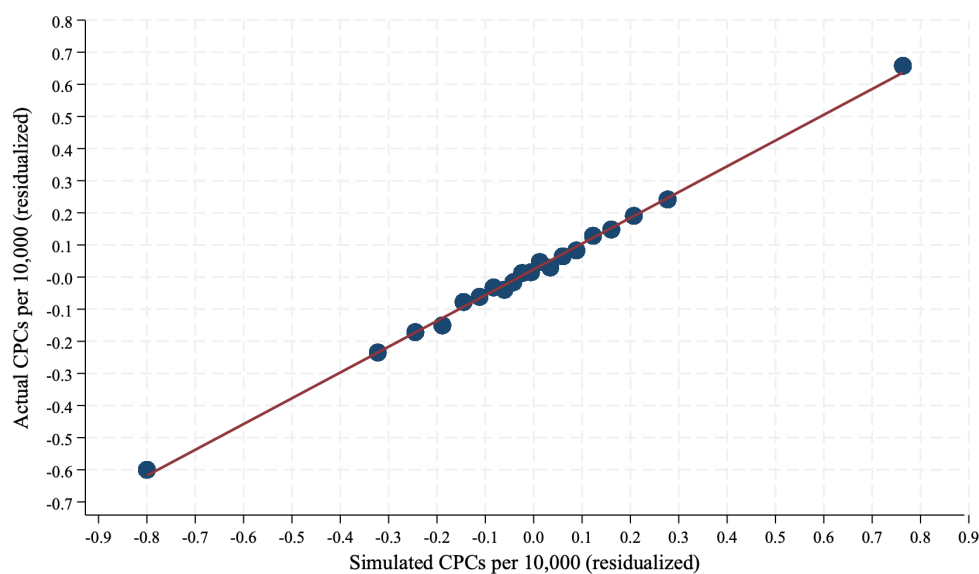
		Standard errors			
	Point est.	Clustered	DK ($\ell = 2$)	DK ($\ell = 3$)	DK ($\ell = 5$)
<i>Panel A: Log abortion rate</i>					
OLS	-0.072	0.029	0.008	0.008	0.008
2SLS	-0.142	0.039	0.023	0.025	0.027
<i>Panel B: Log birth rate</i>					
2SLS	0.018	0.012	0.008	0.008	0.009

Notes: County-year panel of 146 NC and SC counties, 1990–2019 ($N = 4,376$ for log abortion). Each panel reports the baseline 2SLS coefficient with two alternative standard-error estimators. “County-clustered” (paper baseline) clusters at the county level; Driscoll-Kraay standard errors are robust to arbitrary cross-sectional (spatial) and temporal correlation following Driscoll and Kraay (1998), with bandwidth ℓ controlling the number of temporal lags used in the autocorrelation correction. Specifications include county and year fixed effects, the full lagged-control set, and $\sqrt{\text{pop}}$ weights. The coefficient and the 2SLS point estimate are identical across rows by construction; only the standard errors differ.

E.1 First Stage Diagnostics

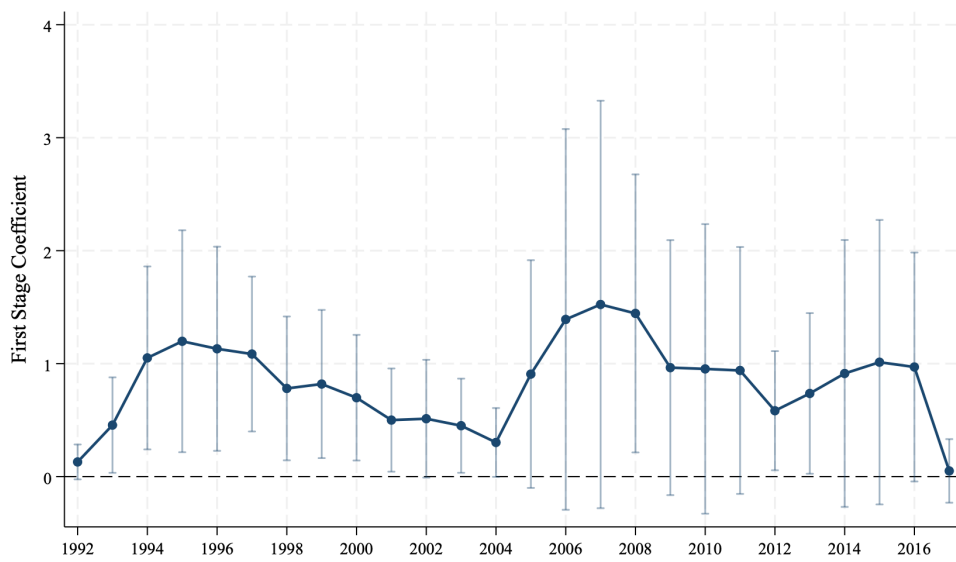
Figure A2 plots actual against simulated CPC intensity after residualizing on county and year fixed effects, confirming a strong positive relationship. Figure A3 reports rolling 5-year-window first-stage coefficients across the sample period. The instrument is strongest in the 1990s and early 2000s, when CPC expansion was most active; coefficients weaken and confidence intervals widen in post-2010 windows as CPC entry slows and the remaining variation becomes less informative. This temporal concentration is consistent with the pre-2003 subsample estimate (-0.142 , $F = 12.6$), which is virtually identical to the full-sample coefficient despite having only half the observations and a weaker first stage, and supports interpreting the main results as primarily identified from the earlier expansion period.

Figure A2: First Stage: Actual vs. Simulated CPC Intensity



Notes: County-year panel of 146 NC and SC counties, 1990–2019. Binscatter of actual CPCs per 10,000 women (vertical axis) against simulated CPCs per 10,000 women (horizontal axis). Both variables are residualized on county fixed effects, year fixed effects, and the full lagged-control set before binning. Twenty equal-mass bins; the solid line is the linear fit through the bin means. The slope visualizes the first-stage relationship; the linear-fit slope across all 4,376 county-years is reported on the figure.

Figure A3: Rolling-Window First-Stage Coefficients



Notes: County-year panel of 146 NC and SC counties. Each point plots the first-stage coefficient from a 2SLS specification estimated on a 5-year rolling window ending in the year on the horizontal axis (so the point at 1995 uses 1991–1995, etc.); the regression is actual CPCs per 10,000 women on simulated CPCs per 10,000 women, with county and year fixed effects and the full lagged-control set. Vertical bars are 95% confidence intervals from standard errors clustered at the county level. The instrument is strongest in the middle of the panel; modest variation reflects sample-size differences and the differential timing of CPC openings.

Table A9: Shuffled Instrument Permutation Test

	First-stage F	Reduced-form $\hat{\pi}$
Actual instrument	196.1	-0.1108 (0.0330)
Shuffled median	0.664	-0.0035
Shuffled mean	1.638	-0.0027
Shuffled SD		0.0270
Shuffled 95th pctile	6.864	
Permutation p -value	0.0000	0.0000

Notes: County-year panel, 1990–2019. 500 permutations randomly reassign county labels to the simulated CPC intensity series, preserving each county’s own time series and the overall cross-sectional distribution. For each permutation the full 2SLS pipeline is re-estimated and the first-stage F and reduced-form coefficient are recorded. “Actual” columns report the values from the un-shuffled instrument; “Permutation median” is the median across the 500 draws; the permutation p -value is the fraction of shuffled statistics whose magnitude exceeds the actual value. The reduced-form outcome here is the inverse-hyperbolic-sine abortion rate—the transformation applied to all 500 permutations—so the “Actual instrument” coefficient (–0.111) differs slightly from the log-rate reduced form of –0.117 in Table A11; sample, controls, fixed effects, weights, and clustering are identical across the two tables. A small permutation p -value indicates that instrument strength is not an artifact of overfitting county characteristics.

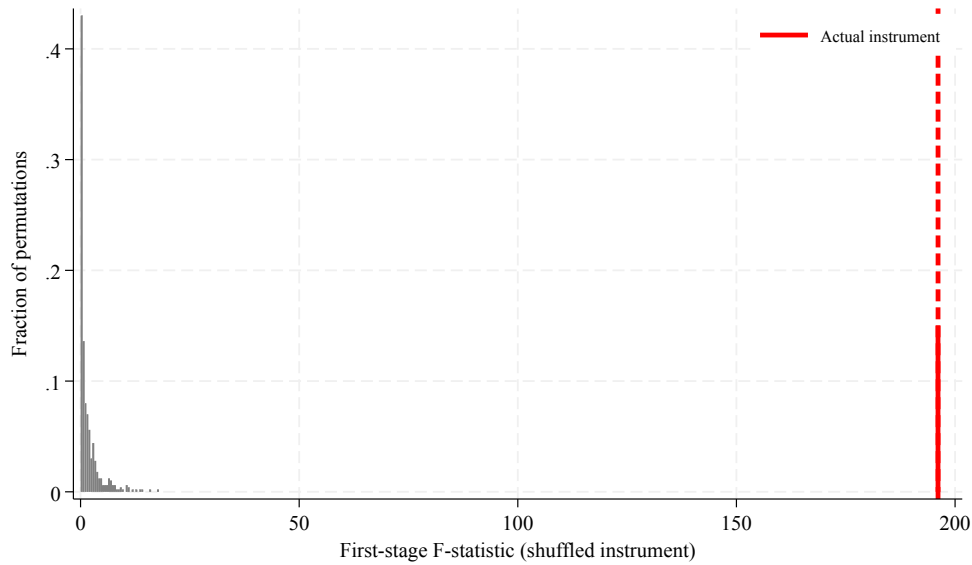


Figure A4: Permutation Distribution of First-Stage F -Statistic

Notes: County-year panel of 146 NC and SC counties, 1990–2019. The histogram is the distribution of the first-stage Kleibergen-Paap F -statistic across 500 random permutations that reassign county labels to the simulated CPC intensity series, preserving each county’s own time-series shape but breaking the link between the simulated intensity and the actual CPC location. Bars are normalized to fractions of the 500 draws. The dashed red line marks the actual first-stage $F = 196.1$ in the un-shuffled instrument. The permutation p -value is the share of shuffled F values exceeding 196.1 (effectively zero), confirming that first-stage strength is not an artifact of overfitting county characteristics.

Table A10: Complier County Baseline Characteristics (1990)

	Full Sample	Top Leverage Q	Bottom 3 Q	p-value
Aggregate Abortion Rate	13.81	12.21	14.33	0.004
Aggregate Birth Rate	50.46	48.94	50.96	0.330
Aggregate Birth/Abortion	4.10	4.22	4.06	0.594
Age 25–29 Abortion Rate	15.82	13.48	16.59	0.017
Age 25–29 Birth Rate	83.31	85.59	82.56	0.680
Age 25–29 Birth/Abortion	6.04	7.02	5.71	0.112
Age 20–24 Abortion Rate	29.19	25.59	30.37	0.050
Age 20–24 Birth Rate	112.74	118.30	110.92	0.332
Age 20–24 Birth/Abortion	4.94	5.44	4.78	0.402

Notes: Leverage quartiles based on SD of simulated CPC intensity across years. Top quartile = counties most affected by the instrument. Birth/Abortion ratio = baseline birth rate / baseline abortion rate. p -values from two-sample t -test with unequal variances.

E.2 Instrument Determinism and Stability

County fixed effects alone explain 72 percent of the instrument's variation; adding year fixed effects raises this to 77 percent, and adding the linear second-stage controls with quadratic terms and interactions brings the total to 82 percent. The remaining 18 percent reflects simulation noise from finite $R = 1,000$ draws and higher-order nonlinear structure in the path-dependent logit. An informal over-identification test re-estimates the baseline 2SLS after dropping each category of time-varying controls from the second stage. The estimates range from -0.109 (dropping economic controls) to -0.177 (dropping demographic controls), all significant at the 1 percent level with first-stage $F > 160$, confirming that no single dimension of $W_{c,t-1}$ drives identification.

Reduced-form estimate. The reduced-form regression of log abortion rates on the simulated instrument confirms the IV result without the first-stage transformation.

Table A11: Reduced-Form Estimates: Outcome on Instrument

	(1) Log Abort	(2) Log Birth	(3) Log Pregnancy
Simulated CPCs per 10,000	-0.117*** (0.034)	0.015 (0.010)	0.012 (0.010)
Observations	4,376	4,380	4,380
Within R^2	0.1271	0.0685	0.1179

Notes: Each column regresses the log outcome directly on the simulated instrument (CPCs per 10,000 women), bypassing the first stage. County and year fixed effects, controls, and $\sqrt{\text{pop}}$ weights. Standard errors clustered at county level. The reduced-form coefficient divided by the first-stage coefficient (0.823) recovers the 2SLS estimate. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

F Welfare: Sufficient-Statistics Bounds and Persuasion Rate

F.1 Implied Persuasion Rate

Table A12 reports the implied DellaVigna and Gentzkow (2010) persuasion rate for CPCs, computed as $f = \Delta y / (e \times y_0)$ where Δy is the aggregate change in the abortion share attributable to CPC expansion, $e = 0.131$ is the CPC exposure rate from Cartwright et al. (2021), and y_0 is the baseline share of pregnancies ending in abortion. Panel A reports aggregate rates for three estimators; Panel B reports age-specific rates using the 2SLS coefficients from Table IV. The exposure rate is drawn from women searching for abortion services online and may overstate exposure among the general pregnant population. The persuasion rate is inversely proportional to e : under a lower exposure of $e = 0.08$, the 2SLS persuasion rate rises to $f = 0.35$; under a higher $e = 0.20$, it falls to $f = 0.14$. The qualitative conclusion—that CPC persuasion rates are comparable to face-to-face canvassing—is robust across this range.

Table A12: Implied CPC Persuasion Rate

	$\hat{\theta}$	Abort share (y_0)	Persuasion rate (f)
<i>Panel A: Aggregate</i>			
2SLS baseline	0.142	0.194	0.211
Poisson CF	0.070	0.194	0.104
Three-lag conservative	0.052	0.194	0.078
<i>Panel B: By Age Group (2SLS)</i>			
Age 10–19	0.162	0.262	0.241
Age 20–24	0.185	0.225	0.274
Age 25–29	0.197	0.156	0.292
Age 30–34	0.151	0.143	0.224
Age 35–44	0.130	0.167	0.193

Notes: Persuasion rate following DellaVigna and Gentzkow (2010): $f = \Delta y / (e \times y_0)$, where Δy is the aggregate change in the abortion share attributable to CPC expansion (Δ CPC intensity = 0.198), $e = 0.131$ is the CPC exposure rate (share of abortion-seeking women who visit a CPC; Cartwright et al., 2021), and y_0 is the baseline share of pregnancies ending in abortion. Panel A: the 2SLS and Poisson CF coefficients are re-estimated from the production instrument and reproduce Table III; the three-lag conservative estimate is Table A5, Column (4). Panel B uses age-specific 2SLS coefficients estimated with the Table IV specification and age-specific abortion shares.

F.2 Structural Identification of the Dollar Scale

The structural calibration in §6.2 parameterizes the surplus distribution as logistic with age-specific parameters (μ_a, σ_a) . The CCP (abortion share of pregnancies) identifies μ_a / σ_a , and the 2SLS treatment effect identifies δ_a / σ_a (the persuasion reach relative to the surplus scale). Both are ratios: the level parameter σ_a —which sets the dollar scale of the surplus distribution—enters only through these ratios and is therefore structurally unidentified from within-sample variation. No specification of the treatment effect model, including quadratic dose-response or heterogeneous treatment effect approaches, can separate δ_a from σ_a without additional information. The externally anchored calibration in Table A13 sets σ_a using the Turnaway Study and teen dropout cost literature; the sufficient-statistics bound in Table A14 avoids specifying σ_a entirely by pric-

ing the treatment effect at the Myers (2017) clinic-distance semi-elasticity. The two approaches bracket the welfare cost per redirected birth at \$59–\$2,554: the lower endpoint is the most conservative clinic-distance bound in Table A14, and the upper endpoint is the high-scenario structural calibration in Table A13.

Table A13: Calibrated Welfare Cost: Age-Specific Logistic Surplus

<i>Panel A: Structural Parameters from Aggregate CCPs</i>					
	10–19	20–24	25–29	30–34	35–44
Abortion share	0.246	0.247	0.146	0.126	0.163
Location ratio	-1.121	-1.115	-1.769	-1.937	-1.638
Persuasion reach	0.3566	0.4145	0.4018	0.2906	0.2555
Persuasion rate	0.244	0.279	0.297	0.228	0.196
Share redirected	15.1%	36.0%	26.3%	13.1%	9.4%
<i>Panel B: Age-Weighted Welfare Cost per Redirected Birth</i>					
Scenario	Target		$\bar{S}_{\text{red}}$	MEB	
Low	\$ 1,500– 5,000		\$ 418	0.405	
Moderate	\$ 3,000–15,000		\$ 1,302	1.263	
High	\$ 8,000–40,000		\$ 2,554	2.477	
Break-even			\$ 1,031	1.000	

Notes: Panel A reports structural parameters identified from aggregate data. Abortion share = abortion rate / (abortion + birth rate), interpreted as the conditional choice probability (Hotz and Miller, 1993). Location ratio = logit of the abortion share, identifying the center of the surplus distribution relative to its scale. Persuasion reach = the CPC’s penetration into the surplus distribution, identified from the age-specific 2SLS treatment effects. Panel B calibrates the dollar scale of the surplus distribution using external evidence on the cost of unwanted births: Low (near-indifferent marginal women), Moderate (teens face dropout risk per Kearney and Levine, 2012; ages 20–24 face Turnaway-level costs per Miller et al., 2023; ages 25–34 adjusted for married births), High (Turnaway-level costs broadly applied). Target = range of age-specific mean surplus among aborters. Surplus follows a logistic distribution (Wolpin, 1984), calibrated following Todd and Wolpin (2006). MEB = welfare cost per dollar of CPC funding (\$ 1,031 per averted abortion, i.e., \$1,000,000 of annual public funding over the 970 averted abortions per year implied by the Table III 2SLS estimate; see Table A14 notes).

F.3 Distance Elasticity Range

The sufficient-statistics conversion $\delta_{\text{CPC}}^{\$} = |\hat{\theta}|/\varepsilon$, where $\delta_{\text{CPC}}^{\$}$ denotes the dollar-equivalent counterpart of the utility shift δ_{CPC} in §4, requires a semi-elasticity of abortion demand per dollar of access cost. Myers (2017) estimates 0.12 percent per additional mile to the nearest provider. Converting miles to dollars requires assumptions about the full cost per mile of distance—including not only gasoline and time but also childcare, work absence, waiting period accommodation, and secrecy costs for minors. Table A14 presents welfare bounds using this clinic-distance elasticity under a range of cost-per-mile assumptions: the welfare cost per redirected birth ranges from \$59 to \$787, with MEB between 0.086 and 0.573. The CPC-specific distance gradient (Online Appendix Table OA.12) is not used as a welfare anchor: its negative sign conditional on the CPC count reflects the CPC–clinic co-location pattern documented in §2 rather than a CPC access cost (Online Appendix §OA.5.3). The clinic-distance bounds price a non-monetary persuasion mechanism using a monetary distance elasticity, likely understating the true welfare significance of the decision while correctly capturing the behavioral magnitude at the margin.

Table A14: Welfare Bounds Under Alternative Cost-of-Distance Assumptions

Cost assumption	\$/mile	$\delta_{CPC}^{\$}$	$\bar{S}$ per redirected birth			MEB
			Convex	Linear	Concave	
Pecuniary only	\$1.50	\$ 177	\$ 59	\$ 89	\$ 118	0.086
Pecuniary + time	\$2.50	\$ 295	\$ 98	\$ 148	\$ 197	0.143
Moderate (+ childcare)	\$4.00	\$ 472	\$ 157	\$ 236	\$ 315	0.229
Full (+ waiting pd.)	\$6.00	\$ 709	\$ 236	\$ 354	\$ 472	0.344
High (+ secrecy)	\$10.00	\$1181	\$ 394	\$ 590	\$ 787	0.573
Break-even (MEB = 1)		\$1031				1.000

Notes: All rows use the Myers (2017) clinic-distance semi-elasticity of abortion demand (0.12% per mile) with varying cost-per-mile assumptions. These bounds price only the access-cost margin of abortion demand: they convert a non-monetary persuasion effect at a monetary distance elasticity and therefore serve as a mechanical lower bound on the welfare cost per redirected birth (§6.2). $\delta_{CPC}^{\$} = |\hat{\theta}|/\epsilon$ is the dollar-equivalent persuasion cost; $\bar{S}$ is the average surplus under convex ($\delta/3$), linear ($\delta/2$), and concave ($2\delta/3$) demand. MEB = welfare cost per dollar of CPC funding. All specifications use the baseline 2SLS estimate from Table III ($|\hat{\theta}| = 0.142$), re-estimated from the production instrument. The break-even figure equals \$1,000,000 of annual public funding divided by the 970 averted abortions per year implied by the baseline estimate; the structural calibration in Table A13 uses the same figure.

F.4 Allcott-Taubinsky Selection Diagnostic

The Allcott-Taubinsky diagnostic cited in §5 tests whether CPC effects concentrate on the women most likely to choose abortion. Across ten age-by-marital-status cells, the cross-cell covariance between baseline abortion propensity and the absolute CPC effect is positive (0.004), the pattern expected under well-targeted persuasion. A separate covariance between baseline LBW rates and the CPC effect on births is negative, indicating CPC-induced births concentrate in cells with healthier baseline outcomes. The table itself appears in Online Appendix §OA.1 (Online Appendix Table OA.3), where the birth-outcome evidence it complements is presented.